

Algebraic Topology of Manifolds II: Characteristic Classes

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About these notes. These notes follow the twelve lectures of the course. They are written to be *read*: each topic opens with the question it answers, and every proof begins with a short paragraph stating the strategy before the argument is carried out step by step. Results imported from outside the course are marked as such, with a reference, rather than proved. The standard reference is Milnor–Stasheff, *Characteristic Classes*.

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Lecture 1

Guiding question. *What is a “continuously varying family of vector spaces” over a space, how do we prove that such a family is genuinely twisted, and which constructions of linear algebra survive the passage from single vector spaces to families?*

1.1 Definitions and first examples

We begin by fixing the stage. A *topological manifold* is a Hausdorff space X in which every point has a neighbourhood homeomorphic to $\mathbb{R}^n$; it is *smooth* when the transition functions between charts are smooth. For a smooth manifold the tangent space $T_x X$ is obtained from paths γ through x , with $\gamma(0) = x$, by taking velocities

$$\left. \frac{d}{dt} \right|_{t=0} \gamma(t).$$

The total space of all tangent spaces is the first and most important example of the structure we are about to define.

Proposition 1.1. *For a smooth n -manifold X , the set $TX = \bigsqcup_{x \in X} T_x X$ is a smooth $2n$ -manifold, the projection $\pi: TX \rightarrow X$ is smooth, each fibre $\pi^{-1}(x) = T_x X$ is a real vector space, and each coordinate chart $U \subset X$ gives a homeomorphism $U \times \mathbb{R}^n \rightarrow \pi^{-1}(U)$ which is linear on each fibre.*

Definition 1.2 (Vector bundle). A *real vector bundle* ξ over a topological space B consists of

1. a topological space $E = E(\xi)$, the *total space*;
2. a continuous surjection $\pi: E \rightarrow B$, the *projection*;
3. a real vector space structure on every fibre $F_b(\xi) := \pi^{-1}(b)$,

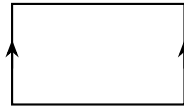
subject to the following *local triviality* condition: every $b \in B$ has a neighbourhood U , an integer $n \geq 0$, and a homeomorphism

$$h: U \times \mathbb{R}^n \longrightarrow \pi^{-1}(U)$$

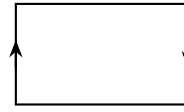
such that $x \mapsto h(b', x)$ is a vector space isomorphism $\mathbb{R}^n \rightarrow F_{b'}(\xi)$ for every $b' \in U$. Such an h is a *local trivialization*. If one may take $U = B$ the bundle is *trivial*; if n is the same for all b we speak of an $\mathbb{R}^n$ -*bundle*, or a bundle of *rank* n .

Example 1.3. Three examples to keep in mind throughout.

1. The cylinder $S^1 \times \mathbb{R}$ is the trivial line bundle over the circle.
2. The *Möbius bundle*: take $[0, 1] \times \mathbb{R}$ and glue $(0, t) \sim (1, -t)$. Locally it is a cylinder; globally it is twisted.
3. TS^2 , an $\mathbb{R}^2$ -bundle over S^2 , which we will prove is non-trivial.



$S^1 \times \mathbb{R}$: edges glued in the same direction



Möbius: $(0, t) \sim (1, -t)$

Definition 1.4. Two vector bundles ξ, η over B are *isomorphic*, written $\xi \cong \eta$, if there is a homeomorphism $f: E(\xi) \rightarrow E(\eta)$ commuting with the projections and restricting to a linear isomorphism $F_b(\xi) \rightarrow F_b(\eta)$ on every fibre.

Definition 1.5. For a smooth submanifold $M \subset \mathbb{R}^n$, the *normal bundle* ν of M is the subspace of $M \times \mathbb{R}^n$ consisting of the pairs (x, v) with $v \perp T_x M$.

Definition 1.6. A (*cross-*)*section* of ξ is a continuous map $s: B \rightarrow E(\xi)$ with $s(b) \in F_b(\xi)$ for every b . It is *nowhere zero* if $s(b) \neq 0$ for all b .

Sections are our main probe. Note three things: a section of a tangent bundle is exactly a vector field; every bundle has a zero section; and a trivial line bundle has a nowhere zero section, obtained by transporting the constant section $b \mapsto (b, 1)$ through a trivialization.

1.2 Sections and non-triviality

Guiding question. *How can one prove that a bundle is non-trivial? The strategy that will serve us for the rest of the course: exhibit something that a trivial bundle must possess, and show the bundle at hand does not possess it.*

Example 1.7 (The canonical line bundle). Write $\mathbb{RP}^n = \{\{\pm x\} \mid x \in S^n\}$ and let $q: S^n \rightarrow \mathbb{RP}^n$ be the quotient map. The *canonical line bundle* $\gamma_n^1 \rightarrow \mathbb{RP}^n$ is the subspace of $\mathbb{RP}^n \times \mathbb{R}^{n+1}$ consisting of pairs $(\{\pm x\}, v)$ with v on the line through x and $-x$, that is, v a multiple of x .

Theorem 1.8. *For every $n \geq 1$ the bundle γ_n^1 is not trivial.*

Proof. Strategy. A trivial line bundle has a nowhere zero section, so it is enough to prove that γ_n^1 has none. Pull a hypothetical section back to the sphere, where the bundle becomes trivial: the section becomes an honest real-valued function, antisymmetric under the antipodal map, and the intermediate value theorem produces a zero.

Step 1. The pull-back of γ_n^1 to S^n is trivial. Consider

$$q^*\gamma_n^1 = \{(x, v) \in S^n \times \mathbb{R}^{n+1} \mid v \in \mathbb{R}x\},$$

$$\Phi: q^*\gamma_n^1 \longrightarrow S^n \times \mathbb{R}, \quad (x, tx) \longmapsto (x, t).$$

The map Φ is a bijection, linear on fibres, and both it and its inverse $(x, t) \mapsto (x, tx)$ are continuous, the former because $t = x \cdot v$ is a polynomial in the coordinates.

Step 2. A section becomes a function. Suppose $s: \mathbb{RP}^n \rightarrow E(\gamma_n^1)$ is a section. Then $x \mapsto s(q(x))$ is a section of $q^*\gamma_n^1$, so by Step 1 there is a continuous $t: S^n \rightarrow \mathbb{R}$ with

$$s(q(x)) = (\{\pm x\}, t(x)x) \quad \text{for all } x \in S^n.$$

Step 3. The function is odd. Since $q(-x) = q(x)$, evaluating the formula of Step 2 at $-x$ gives $t(-x) \cdot (-x) = t(x) \cdot x$, whence

$$t(-x) = -t(x).$$

Step 4. Conclusion. Fix $x \in S^n$ and let α be a path in S^n from x to $-x$, which exists because S^n is path connected for $n \geq 1$. The continuous function $t \circ \alpha$ takes the values $t(x)$ and $-t(x)$ at its endpoints. If $t(x) = 0$ the section already vanishes at $q(x)$; otherwise the two values have opposite signs and the intermediate value theorem gives x_0 on the path with $t(x_0) = 0$, so $s(q(x_0)) = 0$. In either case s is not nowhere zero, and γ_n^1 is not trivial. $\square$

Remark 1.9. For $n = 1$ the total space $E(\gamma_1^1)$ is precisely the Möbius band of Example 1.3: the theorem recovers the familiar fact that the Möbius band is twisted.

The next lemma is the workhorse that lets us verify isomorphisms fibre by fibre, without ever checking continuity of an inverse by hand.

Lemma 1.10. *Let ξ, η be vector bundles over B and let $f: E(\xi) \rightarrow E(\eta)$ be continuous, commuting with the projections, and restricting to a linear isomorphism $F_b(\xi) \rightarrow F_b(\eta)$ for every b . Then f is a homeomorphism and $\xi \cong \eta$.*

Proof. Strategy. Bijectivity is immediate; the whole content is continuity of f^{-1} , which is a local question. In local trivializations f becomes multiplication by an invertible matrix depending continuously on the base point, and Cramer's rule inverts it continuously.

Step 1. f is a bijection. It is injective and surjective on each fibre by hypothesis, and it preserves fibres, so it is a bijection on total spaces.

Step 2. Reduction to a local statement. Continuity of f^{-1} may be checked on the members of an open cover of $E(\eta)$. Fix $b \in B$ and choose a neighbourhood U over which both ξ and η are trivial, with trivializations

$$g: U \times \mathbb{R}^n \rightarrow \pi_\xi^{-1}(U), \quad h: U \times \mathbb{R}^n \rightarrow \pi_\eta^{-1}(U).$$

It suffices to prove that f^{-1} is continuous on $\pi_\eta^{-1}(U)$.

Step 3. f in coordinates. The composite $h^{-1} \circ f \circ g: U \times \mathbb{R}^n \rightarrow U \times \mathbb{R}^n$ preserves the first coordinate and is linear in the second, so it has the form

$$(b', x) \mapsto (b', M(b')x), \quad M(b') \in \text{GL}_n(\mathbb{R}).$$

Each column $M(b')e_i$ is the second component of $h^{-1}fg(b', e_i)$, hence depends continuously on b' ; so all entries of $M(b')$ are continuous functions of b' .

Step 4. Inverting continuously. By Cramer's rule,

$$M(b')^{-1} = \frac{1}{\det M(b')} \text{adj } M(b'),$$

whose entries are rational functions of the entries of $M(b')$ with non-vanishing denominator. Hence $b' \mapsto M(b')^{-1}$ is continuous, and $(b', y) \mapsto (b', M(b')^{-1}y)$ is continuous.

Step 5. Conclusion. On $\pi_\eta^{-1}(U)$ we have $f^{-1} = g \circ (h^{-1}fg)^{-1} \circ h^{-1}$, a composite of continuous maps. Therefore f^{-1} is continuous, f is a homeomorphism, and it is a fibrewise linear isomorphism by hypothesis. $\square$

Definition 1.11. Sections $s_1, \dots, s_n$ of ξ are *nowhere dependent* if the vectors $s_1(b), \dots, s_n(b)$ are linearly independent for every $b \in B$.

Theorem 1.12. An $\mathbb{R}^n$ -bundle ξ is trivial if and only if it admits n nowhere dependent sections.

Proof. Strategy. In both directions the map relating $B \times \mathbb{R}^n$ and E is the obvious one; Lemma 1.10 does the work of checking it is an isomorphism.

Step 1. From sections to a trivialization. Let $s_1, \dots, s_n$ be nowhere dependent and define

$$f: B \times \mathbb{R}^n \longrightarrow E(\xi), \quad f(b, x) = x_1 s_1(b) + \dots + x_n s_n(b).$$

This is continuous, since addition and scalar multiplication are continuous on E and each s_i is continuous; it commutes with the projections; and on the fibre over b it sends the standard basis to the basis $s_1(b), \dots, s_n(b)$ of $F_b(\xi)$, hence is a linear isomorphism there. By Lemma 1.10 it is an isomorphism of bundles, so ξ is trivial.

Step 2. From a trivialization to sections. If $h: B \times \mathbb{R}^n \rightarrow E(\xi)$ is a trivialization, set $s_i(b) = h(b, e_i)$. Each s_i is continuous, and for every b the vectors $s_1(b), \dots, s_n(b)$ are the image of a basis under a linear isomorphism, hence linearly independent. $\square$

Corollary 1.13. *TS^2 is non-trivial.*

Proof. By Theorem 1.12 a trivialization would provide two nowhere dependent vector fields on S^2 , in particular one nowhere zero vector field, contradicting the hairy ball theorem. $\square$

1.3 Constructions

Guiding question. *Which operations of linear algebra — restriction, pull-back, products, direct sums, subspaces, orthogonal complements — make sense for whole families of vector spaces? Essentially all of them, and this flexibility is what makes the theory usable.*

Construction 1.14 (Restriction). If ξ is a bundle over B and $\bar{B} \subset B$, then $\xi|_{\bar{B}}$, with total space $\pi^{-1}(\bar{B})$, is a vector bundle over $\bar{B}$: local trivializations restrict. For smooth $\bar{B} \subset B$ this produces $TB|_{\bar{B}}$.

Construction 1.15 (Pull-back). Given ξ over B and a continuous map $f: B_1 \rightarrow B$, the *induced*, or *pull-back*, bundle $f^*\xi$ over B_1 has total space

$$E_1(f^*\xi) = \{(b, e) \in B_1 \times E(\xi) \mid f(b) = \pi(e)\},$$

topologized as a subspace, with projection $(b, e) \mapsto b$ and the vector space structure of $F_{f(b)}(\xi)$ on the fibre over b . It fits into the square

$$\begin{array}{ccc} E_1(f^*\xi) & \xrightarrow{\hat{f}} & E(\xi) \\ \downarrow & & \downarrow \pi \\ B_1 & \xrightarrow{f} & B \end{array} \quad \hat{f}(b, e) = e.$$

Proposition 1.16. *The pull-back $f^*\xi$ is a vector bundle.*

Proof. Strategy. Only local triviality needs an argument, and a trivialization of ξ near $f(b)$ can simply be pulled back over the preimage.

Step 1. Fix $b \in B_1$ and choose a local trivialization $h: U \times \mathbb{R}^n \rightarrow \pi^{-1}(U)$ of ξ with $f(b) \in U$. Put $U_1 = f^{-1}(U)$, an open neighbourhood of b .

Step 2. Define

$$h_1: U_1 \times \mathbb{R}^n \longrightarrow \pi_1^{-1}(U_1), \quad h_1(b_1, x) = (b_1, h(f(b_1), x)).$$

This lands in E_1 because $\pi(h(f(b_1), x)) = f(b_1)$, and it is continuous because f and h are.

Step 3. It is bijective with continuous inverse $(b_1, e) \mapsto (b_1, \text{pr}_2 h^{-1}(e))$, and for fixed b_1 it is the linear isomorphism $x \mapsto h(f(b_1), x)$ onto $F_{f(b_1)}(\xi)$. Hence h_1 is a local trivialization. $\square$

Definition 1.17. A *bundle map* from η to ξ is a continuous map $g: E(\eta) \rightarrow E(\xi)$ carrying each fibre $F_b(\eta)$ isomorphically onto a fibre $F_{b'}(\xi)$. It induces a map $\bar{g}: B(\eta) \rightarrow B(\xi)$, $b \mapsto b'$, on base spaces, which is continuous.

Lemma 1.18. If $g: E(\eta) \rightarrow E(\xi)$ is a bundle map with base map $\bar{g}$, then $\eta \cong \bar{g}^* \xi$.

Proof. Strategy. There is only one reasonable map to write down; check its fibrewise behaviour and appeal to Lemma 1.10.

Step 1. Define

$$\varphi: E(\eta) \longrightarrow E(\bar{g}^* \xi), \quad \varphi(e) = (\pi_\eta(e), g(e)).$$

This is well defined: by definition of $\bar{g}$ we have $\pi_\xi(g(e)) = \bar{g}(\pi_\eta(e))$, which is exactly the condition defining the pull-back. It is continuous because π_η and g are, and it commutes with the projections to $B(\eta)$.

Step 2. On the fibre over b , the map φ is $e \mapsto (b, g(e))$, and g restricts to an isomorphism $F_b(\eta) \rightarrow F_{\bar{g}(b)}(\xi)$, which is by definition the fibre of $\bar{g}^* \xi$ over b .

Step 3. By Lemma 1.10, φ is an isomorphism of bundles. $\square$

Construction 1.19 (Products and Whitney sums). For ξ_1, ξ_2 over B_1, B_2 the *Cartesian product* $\xi_1 \times \xi_2$ over $B_1 \times B_2$ has total space $E(\xi_1) \times E(\xi_2)$, projection $\pi_1 \times \pi_2$ and fibres $F_{b_1}(\xi_1) \times F_{b_2}(\xi_2)$; products of local trivializations trivialize it. When $B_1 = B_2 = B$, pulling back along the diagonal $d: B \rightarrow B \times B$, $b \mapsto (b, b)$, gives the *Whitney sum*

$$\xi_1 \oplus \xi_2 := d^*(\xi_1 \times \xi_2), \quad F_b(\xi_1 \oplus \xi_2) \cong F_b(\xi_1) \oplus F_b(\xi_2).$$

Definition 1.20. Let ξ, η be bundles over B with $E(\xi) \subset E(\eta)$. We call ξ a *sub-bundle* of η if every $F_b(\xi)$ is a vector subspace of $F_b(\eta)$.

Lemma 1.21. If ξ_1, ξ_2 are sub-bundles of η such that $F_b(\eta) = F_b(\xi_1) \oplus F_b(\xi_2)$ for every b , then $\eta \cong \xi_1 \oplus \xi_2$.

Proof. The map $E(\xi_1 \oplus \xi_2) \rightarrow E(\eta)$ sending $(b, e_1, e_2) \mapsto e_1 + e_2$ is continuous, commutes with the projections, and on the fibre over b is the canonical isomorphism $F_b(\xi_1) \oplus F_b(\xi_2) \rightarrow F_b(\eta)$ provided by the hypothesis. Apply Lemma 1.10. $\square$

1.4 Euclidean vector bundles

Definition 1.22. A *Euclidean vector bundle* is a real vector bundle ξ together with a continuous function $\mu: E(\xi) \rightarrow \mathbb{R}$ which is quadratic and positive definite on every fibre. Polarization turns μ into an inner product on each fibre, varying continuously with the base point; we call it a *Euclidean metric* on ξ , or a *Riemannian metric* when $\xi = TM$.

The space $\mathbb{R}^n$ carries its standard metric, which restricts to any $M \subset \mathbb{R}^n$. On a trivial bundle with a Euclidean metric, Gram–Schmidt — a continuous process — turns the standard frame into n orthonormal nowhere vanishing sections.

Definition 1.23. Let η be a Euclidean bundle of rank n and $\xi \subset \eta$ a sub-bundle of rank k . The *orthogonal complement* $\xi^\perp \subset \eta$ is the union over b of the subspaces $F_b(\xi)^\perp \subset F_b(\eta)$.

Theorem 1.24. $\xi^\perp$ is a vector bundle and $\xi \oplus \xi^\perp \cong \eta$.

Proof. Strategy. Only local triviality of $\xi^\perp$ is at issue, since the second assertion then follows from Lemma 1.21. Near a point, build an orthonormal frame of η whose first k vectors span ξ ; the remaining $n - k$ vectors then trivialize $\xi^\perp$. Producing such a frame uses two standard facts: linear independence is an open condition, and Gram–Schmidt is continuous.

Step 1. Setting up. Fix $b \in B$ and choose a neighbourhood U over which both ξ and η are trivial. Using a trivialization of $\xi|_U$ and Gram–Schmidt, choose sections $s_1, \dots, s_k$ of $\xi|_U$ which are orthonormal at every point of U .

Step 2. Extending the frame at one point. The vectors $s_1(b), \dots, s_k(b)$ are linearly independent in $F_b(\eta)$, so we may complete them to a basis

$$s_1(b), \dots, s_k(b), v_{k+1}, \dots, v_n$$

of $F_b(\eta)$. Using a trivialization of $\eta|_U$, extend the vectors v_j to sections $s'_{k+1}, \dots, s'_n$ of $\eta|_U$ that are constant in that trivialization.

Step 3. Independence persists. In the trivialization of $\eta|_U$ the determinant of the $n \times n$ matrix with columns $s_1, \dots, s_k, s'_{k+1}, \dots, s'_n$ is a continuous function of the base point which is non-zero at b . Hence there is a neighbourhood $V \ni b$, $V \subset U$, on which these n sections are nowhere dependent.

Step 4. Orthonormalizing. Apply Gram–Schmidt pointwise on V to the frame of Step 3. Since the first k vectors are already orthonormal, the process leaves them unchanged and produces sections $s_1, \dots, s_k, s_{k+1}, \dots, s_n$ which form an orthonormal frame of $\eta|_V$ at every point, depend continuously on the base point, and satisfy $\text{span}(s_1, \dots, s_k) = F_{b'}(\xi)$ for $b' \in V$. Consequently $s_{k+1}(b'), \dots, s_n(b')$ is an orthonormal basis of $F_{b'}(\xi)^\perp$.

Step 5. Local triviality of $\xi^\perp$. Define

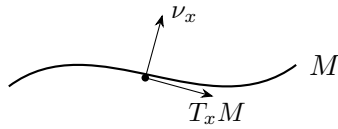
$$V \times \mathbb{R}^{n-k} \longrightarrow \xi^\perp|_V, \quad (b', x) \longmapsto x_1 s_{k+1}(b') + \dots + x_{n-k} s_n(b').$$

It is continuous, fibrewise a linear isomorphism onto $F_{b'}(\xi)^\perp$ by Step 4, and its inverse is $e \mapsto (b', (\langle e, s_{k+j}(b') \rangle)_j)$, continuous because the metric is. Hence $\xi^\perp$ is a vector bundle of rank $n - k$.

Step 6. The sum. By construction $F_b(\eta) = F_b(\xi) \oplus F_b(\xi^\perp)$ for every b , so Lemma 1.21 gives $\xi \oplus \xi^\perp \cong \eta$. $\square$

Example 1.25. If $M \subset N$ are smooth manifolds and N is Riemannian, then TM is a sub-bundle of $TN|_M$ and the *normal bundle of M in N* is $\nu = (TM)^\perp$, so that

$$TN|_M \cong TM \oplus \nu.$$



Definition 1.26. A smooth map $f: M \rightarrow N$ is an *immersion* if $Df_x: T_x M \rightarrow T_{f(x)} N$ is injective for every x . The figure eight is the image of an immersion of S^1 in $\mathbb{R}^2$ which is not an embedding.

Lemma 1.27. For an immersion $f: M \rightarrow N$ with N Riemannian there is a Whitney sum decomposition

$$f^*TN \cong TM \oplus \nu_f,$$

where ν_f is called the normal bundle of the immersion f .

Proof. The differential Df realizes TM as a sub-bundle of f^*TN : it is fibrewise injective by the definition of an immersion, and it is continuous, so its image is a sub-bundle. Pulling back the Riemannian metric of N makes f^*TN Euclidean, and Theorem 1.24 applied to $TM \subset f^*TN$ gives $f^*TN \cong TM \oplus (TM)^\perp$; set $\nu_f = (TM)^\perp$. $\square$

This decomposition is the engine behind many computations to come: it converts geometric information — how M sits inside N — into bundle-theoretic information. In the next lecture we manufacture new bundles from old ones functorially, and meet the first characteristic classes.

Lecture 2

Guiding question. *Linear algebra has functors: duals, homomorphisms, tensor products. Do they act on vector bundles? And can we attach to every bundle algebraic invariants — cohomology classes — that detect twisting?*

2.1 Continuous functors of vector bundles

Definition 2.1. Let Θ be the category whose objects are finite dimensional real vector spaces and whose morphisms are isomorphisms. A functor

$$T: \Theta \times \cdots \times \Theta \longrightarrow \Theta$$

of k variables is *continuous* if $T(f_1, \dots, f_k)$ depends continuously on $f_1, \dots, f_k$, the latter being regarded as matrices.

Theorem 2.2. *Let T be a continuous functor of k variables and let $\xi_1, \dots, \xi_k$ be vector bundles over B . Then there is a vector bundle $T(\xi_1, \dots, \xi_k)$ over B whose fibre over b is $T(F_b(\xi_1), \dots, F_b(\xi_k))$.*

Proof. Strategy. The set and the fibres are prescribed; what has to be produced is a topology. Local trivializations of the ξ_i give candidate charts, and the only thing to check is that the change of chart is continuous — which is exactly the continuity hypothesis on T .

Step 1. The set. Put

$$E = \bigsqcup_{b \in B} T(F_b(\xi_1), \dots, F_b(\xi_k)),$$

with the obvious projection π to B and the vector space structure of $T(\dots)$ on each fibre.

Step 2. Candidate charts. Suppose $U \subset B$ trivializes all the ξ_i simultaneously, with trivializations giving, for $b \in U$, isomorphisms $h_{ib}: \mathbb{R}^{n_i} \rightarrow F_b(\xi_i)$ depending continuously on b . Since T is a functor, applying it to these isomorphisms gives a linear isomorphism

$$T(h_{1b}, \dots, h_{kb}): T(\mathbb{R}^{n_1}, \dots, \mathbb{R}^{n_k}) \longrightarrow T(F_b(\xi_1), \dots, F_b(\xi_k)),$$

and hence a bijection

$$H_U: U \times T(\mathbb{R}^{n_1}, \dots, \mathbb{R}^{n_k}) \longrightarrow \pi^{-1}(U), \quad (b, x) \longmapsto T(h_{1b}, \dots, h_{kb})(x).$$

Step 3. Compatibility of charts. Let U_1, U_2 be two such open sets. On $U_1 \cap U_2$ the composite $H_{U_1}^{-1} \circ H_{U_2}$ is

$$(b, x) \longmapsto \left(b, T(h_{1b}^{(1)-1}h_{1b}^{(2)}, \dots, h_{kb}^{(1)-1}h_{kb}^{(2)})(x) \right),$$

using functoriality. Each $h_{ib}^{(1)-1}h_{ib}^{(2)}$ is the transition isomorphism of ξ_i , which depends continuously on b ; by continuity of T the whole expression depends continuously on (b, x) . The same applies to the inverse.

Step 4. The topology. Declare a subset of E open when its preimage under every H_U is open. By Step 3 the charts H_U are then homeomorphisms onto their images, and they are fibrewise linear isomorphisms by construction. Hence E is a vector bundle with the prescribed fibres. $\square$

Example 2.3. The functor $V \mapsto \text{Hom}(V, \mathbb{R})$ yields the *dual bundle* ξ^* ; two variables give $\xi \otimes \eta$, $\text{Hom}(\xi, \eta)$, $\xi \oplus \eta$, and so on.

Remark 2.4. If ξ is Euclidean, the metric gives an isomorphism $\xi \cong \xi^*$, namely $(b, v) \mapsto (b, \langle v, \cdot \rangle_b)$; it is continuous and fibrewise an isomorphism, so Lemma 1.10 applies. Over a paracompact base a Euclidean metric always exists, by patching local metrics with a partition of unity.

2.2 The Stiefel–Whitney axioms

We now *postulate* a theory of invariants and derive consequences; the construction of the classes will occupy Lectures 4–6. This shortcut lets us prove interesting theorems immediately, which is also how the subject developed historically.

Definition 2.5 (Axioms). To each $\mathbb{R}^n$ -bundle ξ over a base B there are associated cohomology classes

$$w_i(\xi) \in H^i(B; \mathbb{Z}_2), \quad i \geq 0,$$

the *Stiefel–Whitney classes*, subject to:

- (A1) **Rank.** $w_0(\xi) = 1$ and $w_i(\xi) = 0$ for $i > n = \text{rk } \xi$.
- (A2) **Naturality.** If $\hat{f}: \xi \rightarrow \eta$ is a bundle map with base map f , then $w_i(\xi) = f^*w_i(\eta)$; equivalently $w_i(f^*\eta) = f^*w_i(\eta)$.
- (A3) **Whitney product formula.** If ξ, η have the same base,

$$w_k(\xi \oplus \eta) = \sum_{i=0}^k w_i(\xi) \smile w_{k-i}(\eta).$$

(A4) **Normalization.** $w_1(\gamma_1^1) \neq 0$ for the canonical line bundle $\gamma_1^1 \rightarrow \mathbb{RP}^1$.

Proposition 2.6. *If $\xi \cong \eta$ then $w_i(\xi) = w_i(\eta)$ for all i .*

Proof. An isomorphism is a bundle map whose base map is the identity, so (A2) gives $w_i(\xi) = \text{id}^* w_i(\eta) = w_i(\eta)$. $\square$

Proposition 2.7. *If ξ is trivial then $w_i(\xi) = 0$ for every $i > 0$.*

Proof. Strategy. A trivial bundle is pulled back from a bundle over a point, and a point has no cohomology in positive degrees.

Step 1. Write $\xi \cong B \times \mathbb{R}^n$ and let ε' denote the bundle $\text{pt} \times \mathbb{R}^n$ over a point. The map $B \times \mathbb{R}^n \rightarrow \text{pt} \times \mathbb{R}^n$, $(b, x) \mapsto (\text{pt}, x)$, is a bundle map whose base map is the constant map $c: B \rightarrow \text{pt}$.

Step 2. By (A2), $w_i(\xi) = c^* w_i(\varepsilon')$. Since $H^i(\text{pt}; \mathbb{Z}_2) = 0$ for $i > 0$, we have $w_i(\varepsilon') = 0$ for $i > 0$, hence $w_i(\xi) = 0$. $\square$

Proposition 2.8 (Stability). *If ε is a trivial bundle then $w_i(\varepsilon \oplus \eta) = w_i(\eta)$.*

Proof. By Proposition 2.7, $w_0(\varepsilon) = 1$ and $w_j(\varepsilon) = 0$ for $j > 0$; the Whitney formula (A3) then reduces to the single term $w_0(\varepsilon) \smile w_i(\eta) = w_i(\eta)$. $\square$

Proposition 2.9. *If a Euclidean $\mathbb{R}^n$ -bundle ξ admits k nowhere dependent sections, then*

$$w_n(\xi) = w_{n-1}(\xi) = \cdots = w_{n-k+1}(\xi) = 0.$$

Proof. Strategy. The sections span a trivial sub-bundle; splitting it off drops the rank by k , and the rank axiom does the rest.

Step 1. Let $s_1, \dots, s_k$ be the sections and let $\varepsilon^k \subset \xi$ be the sub-bundle whose fibre over b is $\text{span}(s_1(b), \dots, s_k(b))$. It is a sub-bundle of rank k , trivialized by $(b, x) \mapsto \sum x_i s_i(b)$ as in Theorem 1.12.

Step 2. Let $\eta = (\varepsilon^k)^\perp$, a bundle of rank $n - k$ with $\varepsilon^k \oplus \eta \cong \xi$ by Theorem 1.24.

Step 3. By Proposition 2.8, $w_i(\xi) = w_i(\varepsilon^k \oplus \eta) = w_i(\eta)$, and by (A1) applied to η this vanishes for $i > n - k$. $\square$

Remark 2.10. Read contrapositively: if $w_{n-k}(TM) \neq 0$ then M admits at most k everywhere independent vector fields. Stiefel–Whitney classes are *obstructions to finding independent sections*, and this is the historical origin of the theory.

2.3 The total class and Whitney duality

Let $H^\Pi(B; \mathbb{Z}_2)$ denote the ring of formal series $a = a_0 + a_1 + a_2 + \cdots$ with $a_i \in H^i(B; \mathbb{Z}_2)$, with the evident sum and product; the product is commutative and associative. The *total Stiefel–Whitney class* of an $\mathbb{R}^n$ -bundle is

$$w(\xi) = 1 + w_1(\xi) + \cdots + w_n(\xi) \in H^\Pi(B; \mathbb{Z}_2),$$

and the Whitney formula (A3) becomes the single identity

$$w(\xi \oplus \eta) = w(\xi) w(\eta).$$

Proposition 2.11. *The series in $H^\Pi(B; \mathbb{Z}_2)$ with leading term 1 form an abelian group under multiplication.*

Proof. Strategy. Solve $w\bar{w} = 1$ degree by degree; each equation determines the next unknown uniquely.

Step 1. Given $w = 1 + w_1 + w_2 + \cdots$, we seek $\bar{w} = 1 + \bar{w}_1 + \bar{w}_2 + \cdots$ with $w\bar{w} = 1$. Comparing terms of degree $k \geq 1$ gives

$$\bar{w}_k + \bar{w}_{k-1}w_1 + \cdots + \bar{w}_1w_{k-1} + w_k = 0.$$

Step 2. This determines $\bar{w}_k$ uniquely in terms of $\bar{w}_1, \dots, \bar{w}_{k-1}$ and $w_1, \dots, w_k$, so by induction $\bar{w}$ exists and is unique. The first values are

$$\bar{w}_1 = w_1, \quad \bar{w}_2 = w_1^2 + w_2, \quad \bar{w}_3 = w_1^3 + w_3.$$

Step 3. Associativity and commutativity are inherited from $H^*(B; \mathbb{Z}_2)$, and the identity element is 1. $\square$

Lemma 2.12 (Whitney duality). *If $M \subset \mathbb{R}^N$ is a smooth submanifold with normal bundle ν , then*

$$w(\nu) = \bar{w}(TM), \quad \text{that is,} \quad w(TM)w(\nu) = 1.$$

Proof. By Example 1.25, $TM \oplus \nu \cong T\mathbb{R}^N|_M$, which is trivial. Proposition 2.7 gives $w(TM \oplus \nu) = 1$, and the Whitney formula turns this into $w(TM)w(\nu) = 1$; now apply Proposition 2.11. $\square$

Example 2.13. The normal bundle of $S^n \subset \mathbb{R}^{n+1}$ is trivial, spanned by the outward unit normal, so $w(\nu) = 1$ and therefore $w(TS^n) = 1$: all Stiefel–Whitney classes of spheres vanish. Yet TS^2 is non-trivial. The classes are powerful, but they are not complete invariants.

2.4 Computations on projective spaces

Recall $H^*(\mathbb{RP}^n; \mathbb{Z}_2) = \mathbb{Z}_2[a]/(a^{n+1})$ with a the generator of $H^1(\mathbb{RP}^n; \mathbb{Z}_2)$.

Example 2.14. $w(\gamma_n^1) = 1 + a$.

Indeed, the inclusion $j: \mathbb{RP}^1 \hookrightarrow \mathbb{RP}^n$ induced by $\mathbb{R}^2 \subset \mathbb{R}^{n+1}$ is covered by a bundle map $\gamma_1^1 \rightarrow \gamma_n^1$, so (A2) gives $w_1(\gamma_1^1) = j^*w_1(\gamma_n^1)$. By (A4) the left side is non-zero, so $w_1(\gamma_n^1) \neq 0$; since $H^1(\mathbb{RP}^n; \mathbb{Z}_2) = \mathbb{Z}_2\{a\}$, this forces $w_1(\gamma_n^1) = a$. The bundle has rank 1, so $w_i = 0$ for $i \geq 2$ by (A1).

Example 2.15. Let $\gamma^\perp$ be the orthogonal complement of γ_n^1 inside the trivial bundle $\mathbb{RP}^n \times \mathbb{R}^{n+1}$. Then

$$w(\gamma^\perp) = 1 + a + a^2 + \cdots + a^n.$$

Indeed $\gamma_n^1 \oplus \gamma^\perp$ is trivial, so $w(\gamma^\perp) = \overline{(1+a)}$, and one checks directly in $\mathbb{Z}_2[a]/(a^{n+1})$ that

$$(1+a)(1+a+\cdots+a^n) = 1+a^{n+1} = 1.$$

Lemma 2.16. *There are isomorphisms*

$$T\mathbb{RP}^n \cong \text{Hom}(\gamma_n^1, \gamma^\perp) \quad \text{and} \quad T\mathbb{RP}^n \oplus \varepsilon^1 \cong \underbrace{\gamma_n^1 \oplus \cdots \oplus \gamma_n^1}_{n+1},$$

where ε^1 is the trivial line bundle.

Proof. Strategy. Describe tangent vectors of $\mathbb{RP}^n$ as antipodal pairs of tangent vectors of S^n , and observe that such a pair is exactly a linear map from the line through x to its orthogonal complement. The second isomorphism then comes from adding the summand $\text{Hom}(\gamma_n^1, \gamma_n^1)$, which is trivial, and using $\gamma^\perp \oplus \gamma_n^1 = \varepsilon^{n+1}$.

Step 1. Tangent vectors upstairs and downstairs. Let $q: S^n \rightarrow \mathbb{RP}^n$ be the quotient map. Since q is a local diffeomorphism, Dq identifies

$$T\mathbb{RP}^n = \{ \{(x, v), (-x, -v)\} \mid x \cdot x = 1, x \cdot v = 0 \},$$

because $Dq(x, v) = Dq(-x, -v)$ and these are the only identifications.

Step 2. From pairs to homomorphisms. Given such a pair, let $L = \mathbb{R}x \subset \mathbb{R}^{n+1}$ be the corresponding line, that is, the fibre of γ_n^1 over $\{\pm x\}$, and define a linear map

$$\ell: L \longrightarrow L^\perp, \quad \ell(x) = v.$$

Replacing (x, v) by $(-x, -v)$ gives the same ℓ , since $\ell(-x) = -v$. Conversely ℓ determines the pair. This assignment is a bijection

$$T\mathbb{RP}^n \longrightarrow \text{Hom}(\gamma_n^1, \gamma^\perp),$$

continuous with continuous inverse in local trivializations, and linear on each fibre; Lemma 1.10 makes it an isomorphism.

Step 3. The trivial summand. The line bundle $\text{Hom}(\gamma_n^1, \gamma_n^1)$ has the nowhere zero section $b \mapsto \text{id}$, hence is trivial by Theorem 1.12: $\text{Hom}(\gamma_n^1, \gamma_n^1) \cong \varepsilon^1$.

Step 4. Adding up. Using Step 2, Step 3 and additivity of Hom in the second variable,

$$T\mathbb{RP}^n \oplus \varepsilon^1 \cong \text{Hom}(\gamma_n^1, \gamma^\perp) \oplus \text{Hom}(\gamma_n^1, \gamma_n^1) \cong \text{Hom}(\gamma_n^1, \gamma^\perp \oplus \gamma_n^1),$$

and the right-hand side is $\text{Hom}(\gamma_n^1, \varepsilon^{n+1})$, since $\gamma^\perp \oplus \gamma_n^1$ is the trivial bundle $\mathbb{RP}^n \times \mathbb{R}^{n+1}$.

Step 5. Splitting the trivial bundle. Finally

$$\text{Hom}(\gamma_n^1, \varepsilon^{n+1}) \cong \bigoplus_{i=1}^{n+1} \text{Hom}(\gamma_n^1, \varepsilon^1) = \bigoplus_{i=1}^{n+1} (\gamma_n^1)^* \cong \bigoplus_{i=1}^{n+1} \gamma_n^1,$$

the last step because a line bundle with a Euclidean metric is isomorphic to its own dual. $\square$

Theorem 2.17. $w(T\mathbb{RP}^n) = (1+a)^{n+1}$, that is, $w_i(T\mathbb{RP}^n) = \binom{n+1}{i} a^i \pmod{2}$.

Proof. By Proposition 2.8 and Lemma 2.16,

$$w(T\mathbb{RP}^n) = w(T\mathbb{RP}^n \oplus \varepsilon^1) = w((\gamma_n^1)^{\oplus(n+1)}) = w(\gamma_n^1)^{n+1} = (1+a)^{n+1},$$

using the Whitney formula and Example 2.14. $\square$

The mod-2 Pascal triangle

$$\begin{array}{ccccccc} & & & & 1 & & \\ & & & 1 & & 1 & \\ & & 1 & & 0 & & 1 \\ & 1 & & 1 & & 1 & & 1 \\ 1 & & 0 & & 0 & & 0 & & 1 \end{array}$$

already gives $w(T\mathbb{RP}^3) = 1$ and $w(T\mathbb{RP}^4) = 1 + a + a^4$.

Corollary 2.18 (Stiefel). *If $\mathbb{RP}^n$ is parallelizable then $n+1$ is a power of 2.*

Proof. Strategy. Parallelizable means $T\mathbb{RP}^n$ trivial, hence $w = 1$. Over $\mathbb{Z}_2$ the Frobenius identity $(1+a)^{2^r} = 1 + a^{2^r}$ lets us factor $(1+a)^{n+1}$ and exhibit a surviving term unless $n+1$ is a power of 2.

Step 1. If $T\mathbb{RP}^n$ is trivial then $w(T\mathbb{RP}^n) = 1$ by Proposition 2.7, so $(1+a)^{n+1} = 1$ in $\mathbb{Z}_2[a]/(a^{n+1})$.

Step 2. Write $n + 1 = 2^r m$ with m odd. Since squaring is additive mod 2,

$$(1 + a)^{2^r} = 1 + a^{2^r},$$

and therefore

$$(1 + a)^{n+1} = (1 + a^{2^r})^m = 1 + m a^{2^r} + \binom{m}{2} a^{2 \cdot 2^r} + \dots$$

Step 3. Suppose $m > 1$. Then $2^r < n + 1$, so $2^r \leq n$ and the class a^{2^r} is non-zero in $\mathbb{Z}_2[a]/(a^{n+1})$. Its coefficient is $m \equiv 1 \pmod{2}$, so $(1 + a)^{n+1} \neq 1$, contradicting Step 1. Hence $m = 1$ and $n + 1 = 2^r$. $\square$

Remark 2.19. The converse is false: the only parallelizable real projective spaces are $\mathbb{RP}^1, \mathbb{RP}^3, \mathbb{RP}^7$, corresponding to $\mathbb{C}$, the quaternions and the octonions. This requires deeper tools (Bott–Milnor, Kervaire, Adams) and is not proved here.

Example 2.20 (Immersion of $\mathbb{RP}^9$). Suppose M^n immerses in $\mathbb{R}^{n+k}$. By Lemma 1.27 the normal bundle ν_f has rank k , and $TM \oplus \nu_f$ is trivial, so that $w_i(\nu_f) = \bar{w}_i(TM)$. Therefore

$$\bar{w}_i(TM) = 0 \quad \text{for } i > k.$$

For $M = \mathbb{RP}^9$: since $10 = 8 + 2$,

$$w(T\mathbb{RP}^9) = (1 + a)^{10} = (1 + a^8)(1 + a^2) = 1 + a^2 + a^8$$

in $\mathbb{Z}_2[a]/(a^{10})$. Inverting, with $b = a^2$ and $b^5 = a^{10} = 0$, the equation $(1 + b + b^4) \bar{w} = 1$ gives successively the coefficients 1, 1, 1, 1, 0, so

$$\bar{w}(T\mathbb{RP}^9) = 1 + a^2 + a^4 + a^6.$$

The top non-zero term sits in degree 6, so $k \geq 6$: the space $\mathbb{RP}^9$ cannot be immersed in $\mathbb{R}^{9+k}$ for $k < 6$.

Example 2.21 (Immersion of $\mathbb{RP}^{2^r}$). Let $n = 2^r$. Then

$$w(T\mathbb{RP}^n) = (1 + a)^{2^r+1} = (1 + a^{2^r})(1 + a) = 1 + a + a^n$$

in $\mathbb{Z}_2[a]/(a^{n+1})$, and one checks

$$\bar{w}(T\mathbb{RP}^n) = 1 + a + a^2 + \dots + a^{n-1},$$

since $(1+a)(1+a+\dots+a^{n-1}) = 1+a^n$ telescopes and $a^n(1+a+\dots+a^{n-1}) = a^n$ modulo a^{n+1} , so the two extra terms cancel. The top non-zero term is a^{n-1} , so any immersion of $\mathbb{RP}^{2^r}$ needs $k \geq n - 1$, that is, an ambient $\mathbb{R}^{2n-1}$. This is exactly the bound of Whitney's immersion theorem, so projective spaces of 2-power dimension are the worst case.

2.5 Stiefel–Whitney numbers

Guiding question. *Classes live in the cohomology of M , which makes them awkward to compare between different manifolds. Can we extract from them plain numbers, and what do those numbers see?*

Definition 2.22. Let M be a closed smooth n -manifold with fundamental class $\mu_M \in H_n(M; \mathbb{Z}_2)$. For every partition $r_1 + 2r_2 + \cdots + nr_n = n$ the associated *Stiefel–Whitney number* is

$$w_1^{r_1} \cdots w_n^{r_n}[M] := \langle w_1(TM)^{r_1} \smile \cdots \smile w_n(TM)^{r_n}, \mu_M \rangle \in \mathbb{Z}_2.$$

Two n -manifolds *have the same Stiefel–Whitney numbers* if these agree for every such $(r_1, \dots, r_n)$.

Example 2.23. For n even, $w_n(T\mathbb{RP}^n) = \binom{n+1}{n}a^n = (n+1)a^n \neq 0$ and likewise $w_1(T\mathbb{RP}^n)^n = (n+1)^na^n \neq 0$, so $\mathbb{RP}^n$ has non-zero Stiefel–Whitney numbers. For n odd they all vanish, as we verify at the start of the next lecture — and this is no accident: odd projective spaces are boundaries.

In Lecture 3 we prove that Stiefel–Whitney numbers vanish on boundaries, and state Thom’s theorem that they classify closed manifolds up to unoriented cobordism.

Lecture 3

Guiding question. *Stiefel–Whitney numbers are computable invariants of closed manifolds. Which equivalence relation do they detect? And is there a universal space from which all characteristic classes originate?*

3.1 Stiefel–Whitney numbers and cobordism

Example 3.1. All Stiefel–Whitney numbers of $\mathbb{RP}^{2n-1}$ vanish.

Indeed $w(T\mathbb{RP}^{2n-1}) = (1 + a)^{2n} = (1 + a^2)^n$ contains only terms of even degree, so $w_i(T\mathbb{RP}^{2n-1}) = 0$ whenever i is odd. A monomial $w_1^{r_1} \cdots w_{2n-1}^{r_{2n-1}}$ of total degree $\sum i r_i = 2n - 1$ must have $i r_i$ odd for at least one i , hence both i and r_i odd; that factor is $w_i^{r_i} = 0$.

Theorem 3.2. *If M^n is the boundary of a compact smooth $(n + 1)$ -manifold B , then all Stiefel–Whitney numbers of M vanish.*

Proof. Strategy. The tangent bundle of M is the restriction of the tangent bundle of B , up to a trivial summand, so every characteristic class of M in top degree is restricted from B . On the other hand the fundamental class of M is the boundary of the fundamental class of B , and the evaluation pairing turns ∂ into δ . Exactness of the sequence of the pair kills the result.

Step 1. Classes come from B . Along $M = \partial B$ there is a well-defined outward pointing direction, giving a nowhere zero section of $TB|_M$ transverse to TM , hence a trivial line sub-bundle ε^1 with

$$TB|_M \cong TM \oplus \varepsilon^1.$$

By Proposition 2.8 and naturality, for the inclusion $i: M \hookrightarrow B$,

$$i^* w_j(TB) = w_j(TB|_M) = w_j(TM).$$

Step 2. Fundamental classes. With $\mathbb{Z}_2$ coefficients, $H_{n+1}(B, M; \mathbb{Z}_2)$ contains the fundamental class μ_B of the compact manifold with boundary, and the connecting homomorphism of the pair sends

$$\partial: H_{n+1}(B, M; \mathbb{Z}_2) \longrightarrow H_n(M; \mathbb{Z}_2), \quad \mu_B \longmapsto \mu_M.$$

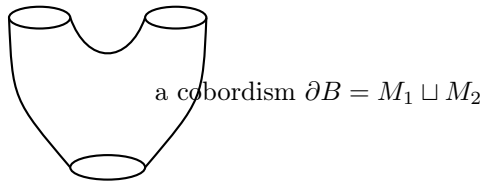
Step 3. Transferring the evaluation. Let $v = w_1(TB)^{r_1} \smile \dots \smile w_n(TB)^{r_n} \in H^n(B; \mathbb{Z}_2)$ with $\sum i r_i = n$. By Step 1, i^*v is the corresponding monomial in the classes of TM . Using Step 2 and the definition of the connecting maps,

$$\langle i^*v, \mu_M \rangle = \langle i^*v, \partial\mu_B \rangle = \langle \delta i^*v, \mu_B \rangle.$$

Step 4. Exactness. The cohomology sequence of the pair (B, M) contains

$$H^n(B; \mathbb{Z}_2) \xrightarrow{i^*} H^n(M; \mathbb{Z}_2) \xrightarrow{\delta} H^{n+1}(B, M; \mathbb{Z}_2),$$

so $\delta \circ i^* = 0$. Hence the expression in Step 3 vanishes, and every Stiefel–Whitney number of M is zero. $\square$



Theorem 3.3 (Thom; quoted without proof). *Conversely, if all Stiefel–Whitney numbers of a closed smooth manifold M vanish, then M is a boundary.*

This converse is genuinely hard; it is proved by the cobordism machinery sketched in Lecture 12. See Milnor–Stasheff, §4 and §18, or Thom’s original paper.

Definition 3.4. Two closed smooth n -manifolds M_1, M_2 are (unoriented) *cobordant* if there is a compact smooth manifold B with $\partial B = M_1 \sqcup M_2$.

Corollary 3.5. *Two closed smooth manifolds are cobordant if and only if their Stiefel–Whitney numbers are equal. In particular the numbers determine the unoriented cobordism class.*

Proof. If they are cobordant, Theorem 3.2 applied to $M_1 \sqcup M_2$ gives $w_I[M_1] + w_I[M_2] = 0$ for every partition I , hence equality since we work mod 2. Conversely, if the numbers agree then all numbers of $M_1 \sqcup M_2$ vanish and Thom’s theorem makes $M_1 \sqcup M_2$ a boundary, that is, a cobordism between M_1 and M_2 . $\square$

3.2 Classifying spaces

Guiding question. *Where do characteristic classes come from? The slogan to aim at: bundles over B are the same thing as homotopy classes of maps*

from B to one fixed universal space, so that every cohomology class of that space produces an invariant of bundles.

Recall that a *fibre bundle* (π, E, B, F) is a continuous surjection $\pi: E \rightarrow B$ with local trivializations $U \times F \cong \pi^{-1}(U)$, and that a *principal G -bundle* is a fibre bundle carrying a G -action which preserves the fibres and is free and transitive on each of them, so that each fibre is homeomorphic to G .

Given a topological group G there is a contractible space EG on which G acts freely; the quotient map

$$\pi: EG \longrightarrow BG = EG/G$$

is the *universal principal G -bundle*, and BG is the *classifying space*. We quote the following.

Theorem 3.6 (quoted). *If M is paracompact and $P \rightarrow M$ is a principal G -bundle, then there is a map $f: M \rightarrow BG$ with $f^*EG \cong P$.*

For vector bundles the relevant group is $GL_n(\mathbb{R})$, or $O(n)$ in the presence of a Euclidean metric: to an $\mathbb{R}^n$ -bundle one associates its *frame bundle*, the principal $GL_n(\mathbb{R})$ -bundle whose fibre over b is the set of frames of $F_b(\xi)$; orthonormal frames give a principal $O(n)$ -bundle. Conversely a principal bundle P recovers the vector bundle $P \times_{GL_n} \mathbb{R}^n$. So the classifying space we want is $BO(n)$, and it has a concrete model: the Grassmannian, with $EO(n)$ modelled on a Stiefel manifold.

3.3 Grassmannians and universal bundles

The motivation is geometric. A curve $C \subset \mathbb{R}^{k+1}$ has a tangent line at each point, giving a map $C \rightarrow \mathbb{RP}^k$; an n -manifold $M \subset \mathbb{R}^{n+k}$ has a tangent n -plane at each point, giving a map to the space of all n -planes.

Definition 3.7. The *Grassmann manifold* $G_n(\mathbb{R}^{n+k})$ is the set of n -dimensional linear subspaces of $\mathbb{R}^{n+k}$. In particular $G_1(\mathbb{R}^{k+1}) = \mathbb{RP}^k$.

Definition 3.8. An *n -frame* in $\mathbb{R}^{n+k}$ is an n -tuple of linearly independent vectors. The *Stiefel manifold* $V_n(\mathbb{R}^{n+k}) \subset (\mathbb{R}^{n+k})^n$ is the set of n -frames, an open subset since it is $\{A \mid \det AA^T \neq 0\}$, the preimage of $\mathbb{R} \setminus \{0\}$ under a continuous map. Let $V_n^0(\mathbb{R}^{n+k}) \subset V_n(\mathbb{R}^{n+k})$ be the subspace of *orthonormal frames*.

We topologize $G_n(\mathbb{R}^{n+k})$ as the quotient of $V_n(\mathbb{R}^{n+k})$ under $q = \text{span}$.

Proposition 3.9. $G_n(\mathbb{R}^{n+k})$ is compact.

Proof. The set $V_n^0(\mathbb{R}^{n+k})$ is closed and bounded in $(\mathbb{R}^{n+k})^n$ — it is cut out by the equations $x_i \cdot x_j = \delta_{ij}$ — hence compact. Gram–Schmidt is a continuous retraction $V_n \rightarrow V_n^0$ commuting with q , so $q|_{V_n^0}: V_n^0 \rightarrow G_n(\mathbb{R}^{n+k})$ is onto. A continuous image of a compact space is compact. $\square$

Theorem 3.10. $G_n(\mathbb{R}^{n+k})$ is a topological manifold of dimension nk , and $X \mapsto X^\perp$ is a homeomorphism $G_n(\mathbb{R}^{n+k}) \cong G_k(\mathbb{R}^{n+k})$.

Proof. Strategy. Hausdorffness is proved by separating two planes with an explicit continuous function, namely the distance to a vector lying in one and not the other. Charts come from the observation that the planes close to a fixed X_0 are exactly the graphs of linear maps $X_0 \rightarrow X_0^\perp$; the identification with $\text{Hom}(X_0, X_0^\perp)$ is the chart.

Step 1. A supply of continuous functions. For $w \in \mathbb{R}^{n+k}$ let $\rho_w(X)$ be the squared distance from w to the plane X . If $(x_1, \dots, x_n)$ is an orthonormal frame of X then

$$\rho_w(X) = w \cdot w - (w \cdot x_1)^2 - \dots - (w \cdot x_n)^2,$$

which is continuous in the frame. Hence $\rho_w \circ q$ is continuous on V_n^0 , and since G_n carries the quotient topology, ρ_w is continuous on $G_n(\mathbb{R}^{n+k})$.

Step 2. Hausdorff. Let $X_1 \neq X_2$ and choose $w \in X_1 \setminus X_2$. Then $\rho_w(X_1) = 0$ while $\rho_w(X_2) > 0$, so the preimages under ρ_w of two disjoint intervals separate X_1 from X_2 .

Step 3. The chart domain. Fix X_0 and split $\mathbb{R}^{n+k} = X_0 \oplus X_0^\perp$. Let

$$U = \{Y \in G_n(\mathbb{R}^{n+k}) \mid \text{proj}_{X_0}|_Y: Y \rightarrow X_0 \text{ is an isomorphism}\}$$

that is, $U = \{Y \mid Y \cap X_0^\perp = 0\}$. This is an open neighbourhood of X_0 : writing $Y = q(y_1, \dots, y_n)$, the condition is that a certain determinant in the frame be non-zero.

Step 4. The chart. For $Y \in U$ every element of Y is determined by its X_0 -component, so Y is the graph

$$Y = \{x + T(Y)x \mid x \in X_0\}$$

of a unique linear map $T(Y): X_0 \rightarrow X_0^\perp$. This gives a bijection

$$T: U \longrightarrow \text{Hom}(X_0, X_0^\perp) \cong \mathbb{R}^{nk}.$$

Step 5. Continuity both ways. Choose an orthonormal basis $x_1, \dots, x_n$ of X_0 . For $Y \in U$ the vectors $y_i = x_i + T(Y)x_i$ form a frame of Y , and conversely any frame of Y close to that of X_0 can be normalized to this form by an invertible linear change, depending continuously on the frame. Hence y_i depends

continuously on Y and vice versa, and $T(Y)$ is determined by the y_i through $T(Y)x_i = y_i - x_i$. So both T and T^{-1} are continuous, and $U \cong \mathbb{R}^{nk}$.

Step 6. The duality homeomorphism. The assignment $Y \mapsto Y^\perp$ is a bijection $G_n(\mathbb{R}^{n+k}) \rightarrow G_k(\mathbb{R}^{n+k})$, and it is continuous: an orthonormal frame of Y can be completed continuously, locally in Y , to an orthonormal frame of $\mathbb{R}^{n+k}$ by the Gram–Schmidt argument of Theorem 1.24, and the last k vectors give a frame of $Y^\perp$. Applying the same to $Y^\perp$ gives continuity of the inverse. $\square$

Definition 3.11. The *tautological bundle* $\gamma^n(\mathbb{R}^{n+k})$ over $G_n(\mathbb{R}^{n+k})$ has total space

$$E = \{(X, \vec{v}) \mid X \in G_n(\mathbb{R}^{n+k}), \vec{v} \in X\}$$

and projection $(X, \vec{v}) \mapsto X$. For $n = 1$ this is $\gamma^1(\mathbb{R}^{k+1}) = \gamma_k^1 \rightarrow \mathbb{RP}^k$.

Lemma 3.12. $\gamma^n(\mathbb{R}^{n+k})$ is locally trivial, hence a vector bundle of rank n .

Proof. Work over the chart domain U of X_0 from Step 3 above. For $Y \in U$ the orthogonal projection $P_Y: Y \rightarrow X_0$ is an isomorphism, so we may define

$$h: U \times X_0 \longrightarrow \pi^{-1}(U), \quad h(Y, x) = (Y, P_Y^{-1}(x)) = (Y, x + T(Y)x).$$

By Step 5 of Theorem 3.10 the map T is continuous, so h is continuous; its inverse is $(Y, v) \mapsto (Y, \text{proj}_{X_0} v)$, also continuous; and for fixed Y it is a linear isomorphism $X_0 \rightarrow Y$. Identifying $X_0 \cong \mathbb{R}^n$ gives a local trivialization. $\square$

Definition 3.13. For a smooth $M^n \subset \mathbb{R}^{n+k}$, the *generalized Gauss map* is

$$\bar{g}: M \longrightarrow G_n(\mathbb{R}^{n+k}), \quad x \longmapsto T_x M,$$

the base map of the bundle map $g: TM \rightarrow \gamma^n(\mathbb{R}^{n+k})$, $(x, v) \mapsto (T_x M, v)$.

So tangent bundles of submanifolds of Euclidean space are pulled back from tautological bundles. The following says this is neither an accident nor a restriction.

Lemma 3.14. Let ξ be an $\mathbb{R}^n$ -bundle over a compact base B . Then for k large enough there is a bundle map $\xi \rightarrow \gamma^n(\mathbb{R}^{n+k})$.

Proof. Strategy. A bundle map into a tautological bundle is the same thing as a map from the total space to a Euclidean space, linear and injective on each fibre; the plane component of the bundle map is then forced. Such a map is assembled from finitely many local trivializations, damped by bump functions so that they extend by zero, and stacked into a single large Euclidean space.

Step 1. Reduction. Suppose we have a continuous $\hat{f}: E(\xi) \rightarrow \mathbb{R}^m$ which is linear and injective on every fibre. Then

$$f(e) = \left(\hat{f}(F_{\pi(e)}(\xi)), \hat{f}(e) \right)$$

defines a map $E(\xi) \rightarrow E(\gamma^n(\mathbb{R}^m))$: the first entry is an n -plane in $\mathbb{R}^m$ because $\hat{f}$ is injective and linear on the fibre, and the second lies in it. It is fibrewise a linear isomorphism onto that plane. Its continuity follows from the continuity of $\hat{f}$ together with the local description of the chart in Lemma 3.12.

Step 2. Bump functions. By compactness choose a finite trivializing cover $U_1, \dots, U_r$ of B , sets $W_i \subset \overline{W_i} \subset V_i \subset \overline{V_i} \subset U_i$ with $\{W_i\}$ still covering B , and continuous $\lambda_i: B \rightarrow [0, 1]$ with $\lambda_i = 1$ on W_i and $\lambda_i = 0$ outside V_i .

Step 3. Damped local coordinates. Each trivialization $h_i: U_i \times \mathbb{R}^n \rightarrow \pi^{-1}(U_i)$ gives $\tilde{h}_i = \text{proj}_{\mathbb{R}^n} \circ h_i^{-1}: \pi^{-1}(U_i) \rightarrow \mathbb{R}^n$, linear and injective on each fibre over U_i . Define

$$h'_i: E(\xi) \longrightarrow \mathbb{R}^n, \quad h'_i(e) = \lambda_i(\pi(e)) \cdot \tilde{h}_i(e)$$

for $\pi(e) \in U_i$, and $h'_i(e) = 0$ otherwise. This is continuous: the two definitions agree on the overlap $U_i \setminus \overline{V_i}$, where $\lambda_i = 0$.

Step 4. Stacking. Set $m = rn$ and

$$\hat{f} = (h'_1, \dots, h'_r): E(\xi) \longrightarrow \underbrace{\mathbb{R}^n \oplus \dots \oplus \mathbb{R}^n}_r = \mathbb{R}^m.$$

It is continuous and linear on fibres. It is injective on the fibre over b : some W_i contains b , so $\lambda_i(b) = 1$ and the i -th component is $\tilde{h}_i$, already injective there.

Step 5. By Step 1 this $\hat{f}$ produces a bundle map $\xi \rightarrow \gamma^n(\mathbb{R}^m)$, and $m = rn$ is of the form $n + k$ with $k = (r - 1)n$. $\square$

To remove the dependence on k , let $\mathbb{R}^\infty = \bigcup_m \mathbb{R}^m$ carry the direct limit topology and define the *infinite Grassmannian*

$$G_n = G_n(\mathbb{R}^\infty) = \varinjlim_k G_n(\mathbb{R}^{n+k}),$$

with tautological bundle γ^n defined by the same formula. This is the *universal n -plane bundle*. In the next lecture we prove that every $\mathbb{R}^n$ -bundle over a paracompact base admits a bundle map to γ^n , unique up to homotopy, so that

$$\{\mathbb{R}^n\text{-bundles over } B\} / \cong \longleftrightarrow [B, G_n],$$

and characteristic classes are then simply elements of $H^*(G_n)$ pulled back along classifying maps.

Lecture 4

Guiding question. *For which base spaces does the slogan “bundles are the same as maps to the Grassmannian” hold, and what does the Grassmannian look like as a space? The technical hypothesis is paracompactness; the geometric answer is that the Grassmannian is a CW complex with exactly one cell for each Schubert symbol.*

4.1 Paracompactness

Definition 4.1. A space is *paracompact* if every open cover admits a locally finite refinement, that is, a refinement such that every point has a neighbourhood meeting only finitely many of its members.

Theorem 4.2 (quoted). *Every metric space is paracompact and Hausdorff. A regular space which is a countable union of compact subsets is paracompact and Hausdorff. Consequently the direct limit of a sequence of compact Hausdorff spaces is paracompact.*

Remark 4.3. The infinite Grassmannian $G_n = \varinjlim_k G_n(\mathbb{R}^{n+k})$ is therefore paracompact, being a direct limit of compact spaces. All bases we shall meet — manifolds, CW complexes, Grassmannians — are paracompact.

The following technical lemma replaces an arbitrary trivializing cover by a countable one. It is the device that makes the classification theorem work for non-compact bases.

Lemma 4.4. *Let ξ be a vector bundle over a paracompact Hausdorff base B . Then there is a countable open cover $U_1, U_2, \dots$ of B such that $\xi|_{U_k}$ is trivial for every k .*

Proof. Strategy. Start from a locally finite trivializing cover $\{V_\alpha\}$ with subordinate bump functions λ_α . For each *finite* index set S collect the points where the functions indexed by S dominate all the others; this produces an open set contained in $\bigcap_{\alpha \in S} V_\alpha$, hence trivializing. Two different sets of the

same cardinality give disjoint pieces, so we may take their union and index the resulting cover by the cardinality $k = |S|$ alone.

Step 1. Set-up. Choose a locally finite open cover $\{V_\alpha\}$ of B such that $\xi|_{V_\alpha}$ is trivial, and continuous functions $\lambda_\alpha: B \rightarrow [0, 1]$ with $\lambda_\alpha = 0$ outside V_α and with the open sets $W_\alpha = \lambda_\alpha^{-1}((0, 1])$ still covering B . Such data exists by paracompactness.

Step 2. The sets $U(S)$. For a finite non-empty set S of indices put

$$U(S) = \left\{ b \in B \mid \min_{\alpha \in S} \lambda_\alpha(b) > \max_{\beta \notin S} \lambda_\beta(b) \right\}.$$

Near any point only finitely many λ 's are non-zero, so both the minimum and the maximum are continuous there, and $U(S)$ is open.

Step 3. $U(S)$ is trivializing. If $b \in U(S)$ and $\alpha \in S$ then $\lambda_\alpha(b) > 0$, so $b \in V_\alpha$. Hence

$$U(S) \subset \bigcap_{\alpha \in S} V_\alpha \subset V_\alpha,$$

and ξ restricted to $U(S)$ is trivial.

Step 4. Sets of equal size are disjoint. Let $S \neq S'$ with $|S| = |S'| = k$. Choose $\alpha \in S \setminus S'$ and $\alpha' \in S' \setminus S$. If b were in $U(S) \cap U(S')$ we would get $\lambda_\alpha(b) > \lambda_{\alpha'}(b)$ from $b \in U(S)$, since $\alpha \in S$ and $\alpha' \notin S$, and symmetrically $\lambda_{\alpha'}(b) > \lambda_\alpha(b)$ from $b \in U(S')$: a contradiction.

Step 5. The cover. Put

$$U_k = \bigsqcup_{|S|=k} U(S),$$

a disjoint union of open trivializing sets by Steps 3 and 4, hence itself open and trivializing, since a bundle trivial on each member of a disjoint family is trivial on their union. Finally the U_k cover B : given b , let $S = \{\alpha \mid \lambda_\alpha(b) \text{ is maximal}\}$, which is finite and non-empty by local finiteness and Step 1; then $b \in U(S) \subset U_{|S|}$. $\square$

4.2 The universal bundle theorem

Theorem 4.5. *Let ξ be an $\mathbb{R}^n$ -bundle over a paracompact base B . Then*

1. *there exists a bundle map $f: \xi \rightarrow \gamma^n$;*
2. *any two bundle maps $\xi \rightarrow \gamma^n$ are homotopic through bundle maps.*

Consequently the homotopy class of the base map $\bar{f} \in \text{Maps}(B, G_n)$ depends only on ξ , and $\xi \cong \bar{f}^ \gamma^n$.*

Proof. Strategy. Existence repeats the construction of Lemma 3.14, with the countable cover of Lemma 4.4 replacing the finite one; the target is $\mathbb{R}^\infty$ instead

of a fixed $\mathbb{R}^m$. For uniqueness, the naive straight-line homotopy between two fibrewise injective maps $\hat{f}, \hat{g}: E(\xi) \rightarrow \mathbb{R}^\infty$ works whenever their values are never negative multiples of each other. In general one first pushes $\hat{f}$ into the odd coordinates and $\hat{g}$ into the even ones, where the obstruction cannot occur, and chains three such homotopies.

Step 1. Existence. Let $U_1, U_2, \dots$ be as in Lemma 4.4, with trivializations $h_i: U_i \times \mathbb{R}^n \rightarrow \pi^{-1}(U_i)$ and bump functions λ_i whose positivity sets cover B . Setting $\tilde{h}_i = \text{proj}_{\mathbb{R}^n} \circ h_i^{-1}$ and $h'_i(e) = \lambda_i(\pi(e)) \tilde{h}_i(e)$, extended by zero, define

$$\hat{f}: E(\xi) \longrightarrow \mathbb{R}^n \oplus \mathbb{R}^n \oplus \dots = \mathbb{R}^\infty, \quad \hat{f}(e) = (h'_1(e), h'_2(e), \dots).$$

Locally only finitely many terms are non-zero, so $\hat{f}$ is continuous; it is linear on fibres, and injective on the fibre over b because $\lambda_i(b) > 0$ for some i . As in Step 1 of Lemma 3.14, $\hat{f}$ determines a bundle map $f: \xi \rightarrow \gamma^n$ by $f(e) = (\hat{f}(F_{\pi(e)}(\xi)), \hat{f}(e))$.

Step 2. The easy homotopy. Suppose $\hat{f}, \hat{g}: E(\xi) \rightarrow \mathbb{R}^\infty$ are continuous, linear and injective on fibres, and satisfy

$$\hat{f}(e) \neq -\lambda^2 \hat{g}(e) \quad \text{for all } e \neq 0 \text{ and all } \lambda \in \mathbb{R}. \quad (*)$$

Then $\hat{h}_t = (1-t)\hat{f} + t\hat{g}$ is fibrewise linear, and injective on fibres: if $\hat{h}_t(v) = 0$ with $v \neq 0$ and $0 < t < 1$ then $\hat{f}(v) = -\frac{t}{1-t}\hat{g}(v)$, contradicting $(*)$; for $t = 0, 1$ it is $\hat{f}$ or $\hat{g}$. So $\hat{h}_t$ gives a homotopy of bundle maps from f to g .

Step 3. Two shift operators. Let $d_1, d_2: \mathbb{R}^\infty \rightarrow \mathbb{R}^\infty$ be the linear injections

$$d_1(e_i) = e_{2i-1}, \quad d_2(e_i) = e_{2i}.$$

Being injective and linear, each induces a bundle map $\gamma^n \rightarrow \gamma^n$ sending $(X, v) \mapsto (d_j X, d_j v)$; and $d_j \circ \hat{f}$ is again continuous, linear and fibrewise injective.

Step 4. $\hat{f}$ and $d_1 \hat{f}$ satisfy $()$.* Let $v = \hat{f}(e) \neq 0$ and let k be the index of its first non-zero coordinate. Suppose $d_1 v = -\lambda^2 v$ with $\lambda \neq 0$. If $k = 1$ then $(d_1 v)_1 = v_1$, so $v_1 = -\lambda^2 v_1$ forces $v_1 = 0$, a contradiction. If $k > 1$ then the first non-zero coordinate of $d_1 v$ sits in position $2k - 1 > k$, so $(d_1 v)_k = 0$, giving $-\lambda^2 v_k = 0$ and again $v_k = 0$. The same argument applies to $\hat{g}$ and $d_2 \hat{g}$, using $2k > k$.

Step 5. $d_1 \hat{f}$ and $d_2 \hat{g}$ satisfy $()$.* The vector $d_1 \hat{f}(e)$ is supported in odd coordinates and $d_2 \hat{g}(e)$ in even ones. If $d_1 \hat{f}(e) = -\lambda^2 d_2 \hat{g}(e)$ then both sides vanish, so $\hat{f}(e) = 0$ and hence $e = 0$.

Step 6. Chaining. By Steps 2, 4 and 5,

$$f \simeq d_1 \circ f \simeq d_2 \circ g \simeq g$$

through bundle maps, which proves (2).

Step 7. Conclusion. A bundle map $f: \xi \rightarrow \gamma^n$ gives $\xi \cong \bar{f}^* \gamma^n$ by Lemma 1.18, and by (2) the class $[\bar{f}]$ is well defined. $\square$

Remark 4.6. The converse also holds: homotopic maps $B \rightarrow G_n$ pull γ^n back to isomorphic bundles, so

$$\{\mathbb{R}^n\text{-bundles over } B\} / \cong \longleftrightarrow [B, G_n].$$

This gives a clean definition of characteristic classes. For $c \in H^i(G_n)$ and a bundle ξ with classifying map f_ξ , put

$$c(\xi) := f_\xi^* c \in H^i(B).$$

Naturality is automatic: if $\eta = g^* \xi$ then $f_\eta \simeq f_\xi \circ g$, so $c(\eta) = g^* c(\xi)$.

$$\begin{array}{ccccc} \eta & \longrightarrow & \xi & \longrightarrow & \gamma^n \\ \downarrow & & \downarrow & & \downarrow \\ B_1 & \xrightarrow{g} & B_2 & \xrightarrow{f_\xi} & G_n \\ & \searrow f_\eta & & & \end{array}$$

The whole theory therefore reduces to one computation: what is $H^*(G_n)$?

4.3 A cell structure on the Grassmannian

Definition 4.7. A *CW complex* is a Hausdorff space K partitioned into disjoint open cells $\{e_\alpha\}$ such that

1. each e_α carries a characteristic map $f: D^{n(\alpha)} \rightarrow K$ taking the open disc homeomorphically onto e_α ;
2. $\bar{e}_\alpha \setminus e_\alpha$ is contained in a finite union of cells of strictly lower dimension;
3. every point lies in a finite subcomplex;
4. K carries the direct limit topology of its finite subcomplexes.

Fix the standard flag $\mathbb{R}^0 \subset \mathbb{R}^1 \subset \dots \subset \mathbb{R}^m$, where $\mathbb{R}^k$ is spanned by the first k coordinates, and let $X \subset \mathbb{R}^m$ be an n -plane. Consider the sequence of integers

$$0 = \dim(X \cap \mathbb{R}^0) \leq \dim(X \cap \mathbb{R}^1) \leq \dots \leq \dim(X \cap \mathbb{R}^m) = n.$$

Claim 4.8. *Consecutive terms differ by at most 1.*

Proof. The sequence

$$0 \longrightarrow X \cap \mathbb{R}^{k-1} \longrightarrow X \cap \mathbb{R}^k \xrightarrow{x \mapsto x_k} \mathbb{R}$$

is exact: the kernel of the k -th coordinate function on $X \cap \mathbb{R}^k$ is exactly $X \cap \mathbb{R}^{k-1}$. Hence

$$\dim(X \cap \mathbb{R}^k) = \dim(X \cap \mathbb{R}^{k-1}) + \text{rk}(x \mapsto x_k) \leq \dim(X \cap \mathbb{R}^{k-1}) + 1.$$

□

So the sequence increases by exactly 1 at n places, the *jumps*

$$1 \leq \sigma_1 < \sigma_2 < \cdots < \sigma_n \leq m,$$

called the *Schubert symbol* of X . Define

$$e(\sigma) = \{X \in G_n(\mathbb{R}^m) \mid \dim(X \cap \mathbb{R}^{\sigma_i}) = i \text{ and } \dim(X \cap \mathbb{R}^{\sigma_{i-1}}) = i - 1, \\ \text{for } i = 1, \dots, n\}.$$

Every X lies in exactly one $e(\sigma)$, and there are $\binom{m}{n}$ symbols.

Claim 4.9. *Let $H^k \subset \mathbb{R}^k \subset \mathbb{R}^m$ be the half-space $\{x \in \mathbb{R}^k \mid x_k > 0\}$. Then $X \in e(\sigma)$ if and only if X has a basis $x_1, \dots, x_n$ with $x_i \in H^{\sigma_i}$; moreover such a basis can be chosen orthonormal, and then it is unique.*

Proof. Strategy. One direction is a rank computation: a vector whose last non-zero coordinate is in position σ_i forces the dimension to jump there. The other direction constructs the basis one vector at a time, taking at each jump the unique unit vector orthogonal to the previous ones and normalized to have positive σ_i -th coordinate.

Step 1. Sufficiency. Suppose X has a basis with $x_i \in H^{\sigma_i}$. Then $x_1, \dots, x_i \in \mathbb{R}^{\sigma_i}$, so $\dim(X \cap \mathbb{R}^{\sigma_i}) \geq i$; and the σ_i -th coordinate of x_i is non-zero while $x_1, \dots, x_{i-1} \in \mathbb{R}^{\sigma_{i-1}} \subset \mathbb{R}^{\sigma_i-1}$, so the map $X \cap \mathbb{R}^{\sigma_i} \rightarrow \mathbb{R}, x \mapsto x_{\sigma_i}$, has rank 1 and $\dim(X \cap \mathbb{R}^{\sigma_i}) > \dim(X \cap \mathbb{R}^{\sigma_i-1})$. Counting the n jumps this forces $\dim(X \cap \mathbb{R}^{\sigma_i}) = i$ exactly, so $X \in e(\sigma)$.

Step 2. Necessity, first vector. Let $X \in e(\sigma)$. Then $X \cap \mathbb{R}^{\sigma_1}$ is a line and $X \cap \mathbb{R}^{\sigma_1-1} = 0$, so every non-zero vector of that line has non-zero σ_1 -th coordinate. There is exactly one unit vector x_1 on it with positive σ_1 -th coordinate, and $x_1 \in H^{\sigma_1}$.

Step 3. Necessity, induction. Suppose orthonormal $x_1, \dots, x_{i-1}$ have been found with $x_j \in H^{\sigma_j}$, spanning $X \cap \mathbb{R}^{\sigma_{i-1}}$. The space $X \cap \mathbb{R}^{\sigma_i}$ has dimension i and contains $X \cap \mathbb{R}^{\sigma_{i-1}}$ as a hyperplane, so its orthogonal complement inside $X \cap \mathbb{R}^{\sigma_i}$ is a line. Since $\dim(X \cap \mathbb{R}^{\sigma_i-1}) = i - 1 = \dim(X \cap \mathbb{R}^{\sigma_{i-1}})$, no non-zero vector of that line lies in $\mathbb{R}^{\sigma_{i-1}}$; hence its two unit vectors have non-zero σ_i -th coordinate, and exactly one of them, call it x_i , has it positive. Then $x_i \in H^{\sigma_i}$, and $x_1, \dots, x_i$ is orthonormal.

Step 4. Uniqueness. At each stage the choice was forced: the line was determined by X and the previous vectors, and the sign by the positivity condition. $\square$

In matrix terms: writing a basis of X as the rows of an $n \times m$ matrix and row reducing, $X \in e(\sigma)$ if and only if X is the row space of a unique matrix

in reduced echelon form whose pivots occupy the columns $\sigma_1, \dots, \sigma_n$,

$$\begin{pmatrix} * & \cdots & * & 1 & 0 & \cdots & & & 0 \\ * & \cdots & * & * & * & 1 & 0 & \cdots & 0 \\ \vdots & & & & & & & \ddots & \\ * & \cdots & * & * & * & * & \cdots & 1 & 0 \cdots 0 \end{pmatrix}$$

The number of free entries is

$$d(\sigma) = (\sigma_1 - 1) + (\sigma_2 - 2) + \cdots + (\sigma_n - n),$$

which will be the dimension of the cell. Following Claim 4.9, set

$$e'(\sigma) = V_n^0(\mathbb{R}^m) \cap (H^{\sigma_1} \times \cdots \times H^{\sigma_n}), \quad \bar{e}'(\sigma) = V_n^0(\mathbb{R}^m) \cap (\bar{H}^{\sigma_1} \times \cdots \times \bar{H}^{\sigma_n}),$$

so that $q = \text{span}$ maps $e'(\sigma)$ bijectively onto $e(\sigma)$. In the next lecture we prove that $\bar{e}'(\sigma)$ is a closed cell of dimension $d(\sigma)$ and that these cells assemble into a CW structure.

Lecture 5

Guiding question. We want $H^*(G_n; \mathbb{Z}_2)$, together with generators we can compute with. The plan has four movements: finish the cell structure and count cells; prove the Leray–Hirsch theorem; use projective bundles to obtain a splitting principle; and conclude that $H^*(G_n; \mathbb{Z}_2)$ is a polynomial ring, which finally constructs the Stiefel–Whitney classes and proves they are unique.

5.1 Schubert cells, completed

Recall the notation of Lecture 4: for a symbol $\sigma = (\sigma_1 < \dots < \sigma_n)$ we have $e(\sigma) \subset G_n(\mathbb{R}^m)$, the set $e'(\sigma)$ of orthonormal frames with $x_i \in H^{\sigma_i}$, its closure $\bar{e}'(\sigma)$ where $x_i \in \bar{H}^{\sigma_i}$, and $d(\sigma) = \sum_i (\sigma_i - i)$.

We first record the tool that drives the induction.

Lemma 5.1. For unit vectors $u, v \in \mathbb{R}^m$ let $T(u, v) \in O(m)$ be the rotation carrying u to v and fixing $(\mathbb{R}u + \mathbb{R}v)^\perp$ pointwise. Then $T(u, v)(x)$ depends continuously on (u, v, x) , and if $u, v \in \mathbb{R}^k$ then $T(u, v)x \equiv x \pmod{\mathbb{R}^k}$ for every x .

Proof. On the plane $\mathbb{R}u + \mathbb{R}v$ the rotation is given by an explicit formula in u, v (Gram–Schmidt in that plane followed by the planar rotation through the angle between u and v), and it is the identity on the orthogonal complement; both pieces depend continuously on (u, v) . If $u, v \in \mathbb{R}^k$ then $T(u, v)$ is the identity on $(\mathbb{R}^k)^\perp$ and preserves $\mathbb{R}^k$, so $T(u, v)x - x \in \mathbb{R}^k$. $\square$

Lemma 5.2. $\bar{e}'(\sigma)$ is a closed cell of dimension $d(\sigma)$ with interior $e'(\sigma)$.

Proof. Strategy. Induct on n . For $n = 1$ the set is a closed hemisphere. For the inductive step, observe that a frame in $\bar{e}'(\sigma_1, \dots, \sigma_{n+1})$ is a frame in $\bar{e}'(\sigma_1, \dots, \sigma_n)$ together with one extra unit vector, orthogonal to the previous ones and lying in $\bar{H}^{\sigma_{n+1}}$. Rotations identify the space of possible extra vectors, which varies with the frame, with one fixed closed cell; this exhibits the whole space as a product.

Step 1. Base case $n = 1$. Here $e'(\sigma_1) = \{x \in \mathbb{R}^{\sigma_1} \mid |x| = 1, x_{\sigma_1} \geq 0\}$, a closed hemisphere of S^{σ_1-1} , hence a closed cell of dimension $\sigma_1 - 1 = d(\sigma_1)$, with interior the open hemisphere $e'(\sigma_1)$.

Step 2. The model for the new vector. Fix $\sigma_1 < \dots < \sigma_{n+1}$ and let $b_i \in \mathbb{R}^m$ be the standard basis vector with a 1 in position σ_i , so that $(b_1, \dots, b_n) \in e'(\sigma_1, \dots, \sigma_n)$. Put

$$D = \{u \in \bar{H}^{\sigma_{n+1}} \mid |u| = 1, b_1 \cdot u = \dots = b_n \cdot u = 0\}.$$

The conditions say that u lies in $\mathbb{R}^{\sigma_{n+1}}$, has last coordinate ≥ 0 , and has vanishing coordinates in the n positions $\sigma_1, \dots, \sigma_n$. So D is a closed hemisphere of a sphere of dimension $\sigma_{n+1} - n - 1$, hence a closed cell of dimension $\sigma_{n+1} - n - 1$.

Step 3. Moving a frame to the standard one. Given $(x_1, \dots, x_n) \in e'(\sigma_1, \dots, \sigma_n)$, set

$$T_{(x)} = T(b_n, x_n) \circ \dots \circ T(b_1, x_1) \in O(m).$$

Then $T_{(x)}b_i = x_i$ for every i . Indeed $T(b_j, x_j)$ with $j < i$ fixes b_i , because b_i is orthogonal to both b_j and x_j : orthogonality to x_j holds since $x_j \in \mathbb{R}^{\sigma_j}$ has vanishing coordinate in position $\sigma_i > \sigma_j$. Then $T(b_i, x_i)$ carries b_i to x_i , and the later rotations $T(b_l, x_l)$ with $l > i$ fix x_i for the same reason. The map $(x) \mapsto T_{(x)}$ is continuous by Lemma 5.1.

Step 4. The product decomposition. Define

$$\Psi: e'(\sigma_1, \dots, \sigma_n) \times D \longrightarrow e'(\sigma_1, \dots, \sigma_{n+1})$$

by

$$\Psi((x_1, \dots, x_n), u) = (x_1, \dots, x_n, T_{(x)}u).$$

This lands where claimed. The vector $T_{(x)}u$ is a unit vector; it is orthogonal to each x_i , since $x_i \cdot T_{(x)}u = T_{(x)}b_i \cdot T_{(x)}u = b_i \cdot u = 0$ as $T_{(x)}$ is orthogonal; and it lies in $\bar{H}^{\sigma_{n+1}}$, because by Lemma 5.1 each rotation used moves a vector only within $\mathbb{R}^{\sigma_i} \subset \mathbb{R}^{\sigma_{n+1}}$, so $T_{(x)}u \equiv u \pmod{\mathbb{R}^{\sigma_{n+1}-1}}$ and the σ_{n+1} -th coordinate of $T_{(x)}u$ equals that of u , which is ≥ 0 .

Step 5. Ψ is a homeomorphism. It is continuous by Step 3. It is injective, since the first n entries determine $T_{(x)}$ and hence $u = T_{(x)}^{-1}(T_{(x)}u)$. It is surjective: given $(x_1, \dots, x_{n+1})$ in the target, put $u = T_{(x)}^{-1}x_{n+1}$ and reverse the computations of Step 4. Source and target are compact Hausdorff, so Ψ is a homeomorphism.

Step 6. Conclusion. By induction $e'(\sigma_1, \dots, \sigma_n)$ is a closed cell of dimension $\sum_{i \leq n} (\sigma_i - i)$; by Step 2, D is a closed cell of dimension $\sigma_{n+1} - (n+1)$. A product of closed cells is a closed cell, and dimensions add, giving $d(\sigma_1, \dots, \sigma_{n+1})$. Under Ψ the interiors correspond, since $T_{(x)}u$ has positive σ_{n+1} -th coordinate exactly when u does; hence the interior is $e'(\sigma_1, \dots, \sigma_{n+1})$. $\square$

Lemma 5.3. *The map $q|_{\bar{e}'(\sigma)}: \bar{e}'(\sigma) \rightarrow G_n(\mathbb{R}^m)$ has image $\overline{e(\sigma)}$ and restricts to a homeomorphism $e'(\sigma) \rightarrow e(\sigma)$. In particular it is a characteristic map for the cell $e(\sigma)$.*

Proof. By Claim 4.9 the restriction $q: e'(\sigma) \rightarrow e(\sigma)$ is a bijection; it is continuous, and its inverse assigns to X its unique positive orthonormal basis, which is continuous by the explicit inductive construction in the proof of that claim. Since $\bar{e}'(\sigma)$ is compact and $G_n(\mathbb{R}^m)$ Hausdorff, $q(\bar{e}'(\sigma))$ is compact, hence closed, and it contains $e(\sigma)$, so it contains $\overline{e(\sigma)}$; conversely $\bar{e}'(\sigma)$ is the closure of $e'(\sigma)$, so its image lies in the closure of $e(\sigma)$. $\square$

Theorem 5.4. *The $\binom{m}{n}$ sets $e(\sigma)$ are the cells of a CW structure on $G_n(\mathbb{R}^m)$, with $\dim e(\sigma) = d(\sigma)$. Passing to the limit $m \rightarrow \infty$ gives a CW structure on G_n with one cell for each partition into at most n parts.*

Proof. Strategy. Everything except the boundary condition has been proved. For that condition one shows that a plane in the closure of $e(\sigma)$ has a symbol τ which is termwise smaller, hence a strictly smaller cell dimension unless $\tau = \sigma$.

Step 1. By Lemma 5.2 and Lemma 5.3, each $e(\sigma)$ is an open cell of dimension $d(\sigma)$ with characteristic map $q|_{\bar{e}'(\sigma)}$, and the $e(\sigma)$ partition $G_n(\mathbb{R}^m)$.

Step 2. Boundaries drop dimension. Let $X \in \overline{e(\sigma)} \setminus e(\sigma)$, say $X = q(x_1, \dots, x_n)$ with $(x_i) \in \bar{e}'(\sigma) \setminus e'(\sigma)$. Since $x_i \in \bar{H}^{\sigma_i} \subset \mathbb{R}^{\sigma_i}$, we get $x_1, \dots, x_i \in X \cap \mathbb{R}^{\sigma_i}$ and therefore

$$\dim(X \cap \mathbb{R}^{\sigma_i}) \geq i \quad \text{for every } i.$$

If τ is the symbol of X , this says exactly $\tau_i \leq \sigma_i$ for all i , whence

$$d(\tau) = \sum_i (\tau_i - i) \leq \sum_i (\sigma_i - i) = d(\sigma).$$

Step 3. Strictness. Equality would force $\tau_i = \sigma_i$ for all i , i.e. $X \in e(\sigma)$, which is excluded. Hence $\overline{e(\sigma)} \setminus e(\sigma)$ is contained in the union of cells of strictly smaller dimension, of which there are finitely many.

Step 4. Limit. The inclusions $G_n(\mathbb{R}^m) \subset G_n(\mathbb{R}^{m+1})$ are inclusions of subcomplexes, since a symbol for $\mathbb{R}^m$ is one for $\mathbb{R}^{m+1}$ and the cells match. The limit therefore inherits a CW structure, with one cell for each sequence $1 \leq \sigma_1 < \dots < \sigma_n$, that is, for each partition $0 \leq \sigma_1 - 1 \leq \dots \leq \sigma_n - n$ into at most n parts. $\square$

Corollary 5.5. *For every r ,*

$$\dim_{\mathbb{Z}_2} H^r(G_n; \mathbb{Z}_2) \leq p_n(r) := \#\{\text{partitions of } r \text{ into at most } n \text{ parts}\}.$$

Proof. The cellular cochain complex computing $H^*(G_n; \mathbb{Z}_2)$ has exactly $p_n(r)$ generators in degree r , by Theorem 5.4; cohomology of a complex is a subquotient of the complex. $\square$

Remark 5.6 (quoted). In fact the mod 2 cellular boundary operator vanishes identically, so the inequality above is an equality. We shall not need this: the reverse inequality will come for free from Theorem 5.15 below. See Milnor–Stasheff, §6, for the direct verification that all incidence numbers of Schubert cells are even.

5.2 The Leray–Hirsch theorem

Theorem 5.7 (Leray–Hirsch). *Let $\pi: E \rightarrow B$ be a fibre bundle with fibre F and $i: F \hookrightarrow E$ the inclusion of a fibre. Assume that B admits a finite trivializing cover, that $H^*(F; \Lambda)$ is a free Λ -module of finite rank, and that there is a Λ -linear map*

$$s: H^*(F; \Lambda) \longrightarrow H^*(E; \Lambda) \quad \text{with} \quad i^* \circ s = \text{id}_{H^*(F)}.$$

Then

$$\Phi: H^*(F) \otimes_{\Lambda} H^*(B) \longrightarrow H^*(E), \quad \alpha \otimes \beta \longmapsto s(\alpha) \smile \pi^* \beta,$$

is an isomorphism. In particular $\pi^: H^*(B) \rightarrow H^*(E)$ is injective.*

Proof. Strategy. The map Φ is a homomorphism of $H^*(B)$ -modules and is defined globally, so it can be compared over the members of a cover. Over a trivializing open set Künneth applies, and the matrix of Φ in the Künneth basis is triangular with respect to the degree of the fibre classes, with invertible diagonal blocks; hence Φ is an isomorphism there. Mayer–Vietoris and the five lemma then propagate the statement from a cover to a union, and induction on the size of the cover finishes.

Step 1. Restriction. For open $U \subset B$ let $E_U = \pi^{-1}(U)$ and let Φ_U be defined by the same formula, using the restrictions of $s(\alpha)$ to E_U . Restriction of cohomology classes commutes with Φ , so we obtain maps of long exact sequences whenever we use Mayer–Vietoris.

Step 2. The trivial case. Assume $E_U \cong U \times F$. Choose a homogeneous basis $\alpha_1, \dots, \alpha_N$ of $H^*(F)$ ordered so that $\deg \alpha_1 \geq \dots \geq \deg \alpha_N$. By Künneth,

$$H^*(E_U) \cong H^*(U) \otimes H^*(F),$$

a free $H^*(U)$ -module with basis $1 \otimes \alpha_1, \dots, 1 \otimes \alpha_N$. Write

$$s(\alpha_j)|_{E_U} = \sum_k c_{jk} \otimes \alpha_k, \quad c_{jk} \in H^{\deg \alpha_j - \deg \alpha_k}(U).$$

Since degrees are non-negative, $c_{jk} = 0$ unless $\deg \alpha_j \geq \deg \alpha_k$, so the matrix (c_{jk}) is block triangular for our ordering. Restricting to a fibre kills all classes of positive degree pulled back from U , so $i^*s = \text{id}$ gives, for $\deg \alpha_j = \deg \alpha_k$, that $c_{jk} \in H^0(U)$ has the value δ_{jk} at every point; hence the diagonal blocks are invertible over $H^0(U)$.

Step 3. Invertibility. A block triangular matrix over a commutative ring with invertible diagonal blocks is invertible, because its strictly triangular part is nilpotent (it strictly decreases the finite ordering). Hence (c_{jk}) is invertible over $H^*(U)$, and therefore

$$\Phi_U(\alpha_j \otimes \beta) = s(\alpha_j) \smile \pi^* \beta = \sum_k (c_{jk} \beta) \otimes \alpha_k$$

is an isomorphism of $H^*(U)$ -modules.

Step 4. Mayer–Vietoris. Suppose $U = U' \cup U''$ with the theorem known for U', U'' and $U' \cap U''$. Tensoring the Mayer–Vietoris sequence of (U', U'') with the free module $H^*(F)$ keeps it exact, and Φ maps it to the Mayer–Vietoris sequence of $(E_{U'}, E_{U''})$, commutatively by Step 1. Three out of every four consecutive vertical maps are isomorphisms, so the five lemma gives that Φ_U is an isomorphism.

Step 5. Induction. Let $U_1, \dots, U_r$ be a finite trivializing cover of B . The theorem holds over each U_i by Step 2, hence over $U_1 \cup U_2$ by Step 4, since $U_1 \cap U_2$ is trivializing as well. Assuming it over $U_1 \cup \dots \cup U_j$, apply Step 4 to $U' = U_1 \cup \dots \cup U_j$ and $U'' = U_{j+1}$, whose intersection is $\bigcup_{i \leq j} (U_i \cap U_{j+1})$, a union of j trivializing sets to which the inductive hypothesis applies. After r steps we reach B .

Step 6. Injectivity of π^ .* Taking $\alpha = 1 \in H^0(F)$ and noting $s(1) = 1$, since i^* is injective in degree 0 on the relevant summand, the composite $\beta \mapsto \Phi(1 \otimes \beta) = \pi^* \beta$ is the inclusion of a direct summand, hence injective. $\square$

Remark 5.8. For the infinite Grassmannians we apply the theorem to the compact approximations $G_n(\mathbb{R}^{n+k})$, which do admit finite trivializing covers, and pass to the limit; the cohomology of G_n in each fixed degree agrees with that of $G_n(\mathbb{R}^{n+k})$ for k large, since the CW structures are compatible and no new cells appear in low dimensions.

5.3 Projective bundles and the splitting principle

Definition 5.9. For a real bundle $\pi: E \rightarrow B$ of rank n , the *projective bundle* $p: \mathbb{P}(E) \rightarrow B$ is the fibre bundle whose fibre over b is the space $\mathbb{P}(\pi^{-1}(b)) \cong \mathbb{RP}^{n-1}$ of lines in the fibre.

Lemma 5.10. *For every rank n bundle $E \rightarrow B$ over a base with a finite trivializing cover,*

$$H^*(\mathbb{P}(E); \mathbb{Z}_2) \cong H^*(B; \mathbb{Z}_2) \otimes H^*(\mathbb{RP}^{n-1}; \mathbb{Z}_2)$$

as $H^(B)$ -modules; in particular $p^*: H^*(B; \mathbb{Z}_2) \rightarrow H^*(\mathbb{P}(E); \mathbb{Z}_2)$ is injective.*

Proof. Strategy. Produce the splitting s required by Leray–Hirsch from a canonical line bundle living on $\mathbb{P}(E)$: its classifying map restricts on each fibre to the inclusion $\mathbb{RP}^{n-1} \hookrightarrow \mathbb{RP}^\infty$, which is injective on cohomology in the relevant range.

Step 1. The tautological line bundle. Over the point $(b, \ell) \in \mathbb{P}(E)$, with $\ell \subset \pi^{-1}(b)$ a line, take the line ℓ itself. This defines a line bundle $\gamma \rightarrow \mathbb{P}(E)$; local triviality is inherited from that of E together with the local triviality of γ^1 over $\mathbb{RP}^{n-1}$.

Step 2. Its classifying map. Let $f: \mathbb{P}(E) \rightarrow G_1 = \mathbb{RP}^\infty$ classify γ . On a fibre $F = \mathbb{P}(\pi^{-1}(b)) \cong \mathbb{RP}^{n-1}$ the bundle γ restricts to the canonical line bundle of $\mathbb{RP}^{n-1}$, so $f|_F$ classifies it and is therefore homotopic to the standard inclusion $j: \mathbb{RP}^{n-1} \hookrightarrow \mathbb{RP}^\infty$.

Step 3. The splitting. With $H^*(\mathbb{RP}^\infty; \mathbb{Z}_2) = \mathbb{Z}_2[a]$ and $H^*(\mathbb{RP}^{n-1}; \mathbb{Z}_2) = \mathbb{Z}_2[a]/(a^n)$, the map j^* sends a^k to a^k and is an isomorphism in degrees $\leq n-1$. Define

$$s: H^*(\mathbb{RP}^{n-1}; \mathbb{Z}_2) \rightarrow H^*(\mathbb{P}(E); \mathbb{Z}_2), \quad s(a^k) = f^*(a^k) \quad (0 \leq k \leq n-1),$$

extended linearly. By Step 2, $i_F^* s(a^k) = j^*(a^k) = a^k$, so $i_F^* \circ s = \text{id}$.

Step 4. Apply Theorem 5.7. □

Definition 5.11. A *splitting map* for a rank n bundle $E \rightarrow B$ is a map $f: B' \rightarrow B$ such that $f^*E \cong L_1 \oplus \cdots \oplus L_n$ with all L_i line bundles, and $f^*: H^*(B) \rightarrow H^*(B')$ injective.

Corollary 5.12 (Splitting principle). *Every real vector bundle admits a splitting map.*

Proof. Strategy. Split off one line at a time on the projectivization, where the tautological line bundle is by construction a sub-bundle of the pull-back; the rank drops by one at each stage and injectivity is preserved because a composite of injections is injective.

Step 1. On $\mathbb{P}(E)$ the tautological line bundle γ of Step 1 above is a sub-bundle of p^*E : over (b, ℓ) the fibre of p^*E is $\pi^{-1}(b) \supset \ell$. Choosing a Euclidean metric, Theorem 1.24 gives

$$p^*E \cong \gamma \oplus \gamma^\perp, \quad \text{rk } \gamma^\perp = n-1.$$

Step 2. By Lemma 5.10, p^* is injective on cohomology.

Step 3. Repeat the construction with $\gamma^\perp$ in place of E : form $\mathbb{P}(\gamma^\perp)$, split off a further line, and so on. After $n - 1$ steps the pull-back of E to

$$B' = \mathbb{P}(\cdots \mathbb{P}(\gamma^\perp) \cdots)$$

is a sum of n line bundles, and the composite of the projections is injective on cohomology, being a composite of injective maps. $\square$

Any identity between characteristic classes which holds for sums of line bundles therefore holds in general. This is the workhorse of the subject.

5.4 Cohomology of the Grassmannian

Definition 5.13. A polynomial $p \in \mathbb{Z}_2[a_1, \dots, a_n]$ is *symmetric* if it is invariant under all permutations of the variables. The *elementary symmetric functions* are

$$\sigma_1 = \sum_i a_i, \quad \sigma_2 = \sum_{i < j} a_i a_j, \quad \dots, \quad \sigma_n = a_1 a_2 \cdots a_n.$$

Theorem 5.14 (fundamental theorem of symmetric polynomials, quoted). *The subring of symmetric polynomials in $\mathbb{Z}_2[a_1, \dots, a_n]$ is freely generated by $\sigma_1, \dots, \sigma_n$:*

$$\mathbb{Z}_2[a_1, \dots, a_n]^{S_n} = \mathbb{Z}_2[\sigma_1, \dots, \sigma_n].$$

Recall that $H^*(\mathbb{RP}^\infty; \mathbb{Z}_2) = \mathbb{Z}_2[a]$ and that, by Künneth,

$$H^*((\mathbb{RP}^\infty)^n; \mathbb{Z}_2) = \mathbb{Z}_2[a_1, \dots, a_n]$$

with each a_i of degree 1.

Theorem 5.15. *Let $h_n: (\mathbb{RP}^\infty)^n \rightarrow G_n$ classify $\gamma^1 \oplus \cdots \oplus \gamma^1$, one summand pulled back from each factor. Then*

$$h_n^*: H^*(G_n; \mathbb{Z}_2) \longrightarrow \mathbb{Z}_2[a_1, \dots, a_n]$$

is injective with image the ring of symmetric polynomials. Consequently

$$H^*(G_n; \mathbb{Z}_2) = \mathbb{Z}_2[\tilde{\sigma}_1, \dots, \tilde{\sigma}_n], \quad \tilde{\sigma}_i := (h_n^*)^{-1}(\sigma_i) \in H^i(G_n; \mathbb{Z}_2),$$

and $\dim_{\mathbb{Z}_2} H^r(G_n; \mathbb{Z}_2) = p_n(r)$ for every r .

Proof. Strategy. Injectivity comes from the splitting principle: a splitting map for the universal bundle factors through h_n , and an injective map cannot factor through a non-injective one. Symmetry of the image comes from permuting

the factors, which does not change the bundle being classified. Finally the image is all of the symmetric polynomials by comparing dimensions with the cell count of Corollary 5.5.

Step 1. Injectivity. Let $f: B' \rightarrow G_n$ be a splitting map for γ^n (Corollary 5.12), so $f^*\gamma^n \cong L_1 \oplus \cdots \oplus L_n$ and f^* is injective. Let $g: B' \rightarrow (\mathbb{RP}^\infty)^n$ have i -th component a classifying map for L_i ; then

$$g^*(\gamma^1 \oplus \cdots \oplus \gamma^1) \cong L_1 \oplus \cdots \oplus L_n \cong f^*\gamma^n.$$

So f and $h_n \circ g$ both classify the same bundle over B' , hence are homotopic by Theorem 4.5, giving $f^* = g^* \circ h_n^*$. Since f^* is injective, so is h_n^* .

Step 2. The image is symmetric. A permutation ρ of the factors of $(\mathbb{RP}^\infty)^n$ satisfies $\rho^*(\gamma^1 \oplus \cdots \oplus \gamma^1) \cong \gamma^1 \oplus \cdots \oplus \gamma^1$, since it merely reorders the summands. Hence $h_n \circ \rho \simeq h_n$ and $\rho^*h_n^* = h_n^*$. As ρ^* permutes $a_1, \dots, a_n$, every element of the image is invariant under permutations, so

$$\text{im } h_n^* \subset \mathbb{Z}_2[\sigma_1, \dots, \sigma_n].$$

Step 3. Dimension count. In degree r , the symmetric polynomials have $\mathbb{Z}_2$ -dimension equal to the number of monomials $\sigma_1^{d_1} \cdots \sigma_n^{d_n}$ with $\sum i d_i = r$, that is, to the number $p_n(r)$ of partitions of r into at most n parts (take d_i parts equal to i). By Steps 1 and 2 together with Corollary 5.5,

$$\dim H^r(G_n; \mathbb{Z}_2) \leq p_n(r) \quad \text{and} \quad \dim \text{im } h_n^* = \dim H^r(G_n; \mathbb{Z}_2).$$

Step 4. Surjectivity onto the symmetric part. It remains to produce classes mapping onto $\sigma_1, \dots, \sigma_n$. The mod-2 cellular boundary of the Schubert complex vanishes (quoted above, Milnor–Stasheff §6), so equality holds in Corollary 5.5: $\dim H^r(G_n; \mathbb{Z}_2) = p_n(r)$. Combined with Step 3 this forces $\text{im } h_n^*$ to be the whole degree- r symmetric part, for every r . Hence h_n^* is an isomorphism onto $\mathbb{Z}_2[\sigma_1, \dots, \sigma_n]$ and we may define $\tilde{\sigma}_i$ as its preimage of σ_i . $\square$

Remark 5.16. Step 4 is the one place where we import a computation. It can be avoided: the independent construction of Stiefel–Whitney classes by Steenrod squares in Lecture 7 produces classes $w_i(\gamma^n) \in H^i(G_n; \mathbb{Z}_2)$ with $h_n^*w_i(\gamma^n) = \sigma_i$, which gives the surjectivity of Step 4 directly, and then the equality $\dim H^r = p_n(r)$ becomes a corollary rather than an input. Either route closes the argument; only the order of exposition changes.

5.5 Existence and uniqueness of the SW classes

Theorem 5.17. *There is exactly one family of classes satisfying the axioms of Definition 2.5, namely*

$$w_i(\xi) = f_\xi^*(\tilde{\sigma}_i),$$

where $f_\xi: B \rightarrow G_n$ is a classifying map for ξ .

Proof. Strategy. Existence: three of the four axioms are immediate from the definition, and the fourth — the product formula — is proved in Lecture 6 by computing the effect on cohomology of the map $G_n \times G_{n'} \rightarrow G_{n+n'}$ classifying the direct sum. Uniqueness: naturality reduces everything to the universal bundle, the splitting principle reduces the universal bundle to sums of line bundles, and the product formula reduces sums of line bundles to a single line bundle, where normalization pins the answer down.

Step 1. Rank. $\tilde{\sigma}_0 = 1$ and $H^*(G_n)$ has no generators beyond $\tilde{\sigma}_n$, so $w_0 = 1$ and $w_i = 0$ for $i > n$ by definition.

Step 2. Naturality. If $\eta = g^*\xi$ then $f_\eta \simeq f_\xi \circ g$ by Theorem 4.5, so $w_i(\eta) = f_\eta^* \tilde{\sigma}_i = g^* f_\xi^* \tilde{\sigma}_i = g^* w_i(\xi)$.

Step 3. Normalization. The inclusion $\mathbb{RP}^1 \hookrightarrow \mathbb{RP}^\infty = G_1$ classifies γ_1^1 , and $\tilde{\sigma}_1 = a$ generates $H^1(\mathbb{RP}^\infty; \mathbb{Z}_2)$, whose restriction to $H^1(\mathbb{RP}^1; \mathbb{Z}_2)$ is the generator; hence $w_1(\gamma_1^1) \neq 0$.

Step 4. Product formula. Proved in Theorem 6.2 of the next lecture.

Step 5. Uniqueness. Let w'_i be any family satisfying the axioms. By naturality both families are determined by their values on the universal bundle γ^n , so it suffices to prove $w_i(\gamma^n) = w'_i(\gamma^n)$. Since h_n^* is injective (Theorem 5.15), it suffices to compare after applying h_n^* , that is, to compare the classes of $\gamma^1 \oplus \cdots \oplus \gamma^1$ over $(\mathbb{RP}^\infty)^n$. By the product formula, both are determined by the classes of a single $\gamma^1 \rightarrow \mathbb{RP}^\infty$. There only w_1 is in question, by the rank axiom, and naturality applied to $\gamma_1^1 \rightarrow \gamma^1$ together with the normalization axiom forces $w'_1(\gamma^1)$ to be non-zero, hence equal to the generator a , which is $w_1(\gamma^1)$. $\square$

Lecture 6

Guiding question. *One axiom is still owed: the Whitney product formula. The universal way to prove an identity between characteristic classes is to verify it on universal examples, where cohomology is a polynomial ring and everything reduces to symmetric functions. The same circle of ideas then produces the Chern classes of complex bundles, with integer rather than mod-2 coefficients.*

6.1 Proof of the Whitney product formula

Recall from Lecture 5 that $H^*(G_n; \mathbb{Z}_2) = \mathbb{Z}_2[\tilde{\sigma}_1, \dots, \tilde{\sigma}_n]$, embedded by h_n^* as the symmetric polynomials in $\mathbb{Z}_2[a_1, \dots, a_n]$, and that $w_i(E) = f_E^*(\tilde{\sigma}_i)$.

Write $G = G_n$, $G' = G_{n'}$, $G'' = G_{n+n'}$, and let

$$F: G \times G' \longrightarrow G''$$

classify $p_1^* \gamma^n \oplus p_2^* \gamma^{n'}$, where p_1, p_2 are the two projections.

Lemma 6.1. $F^*(\tilde{\sigma}_i) = \sum_{s=0}^i \tilde{\sigma}'_s \times \tilde{\sigma}''_{i-s}$, where $\tilde{\sigma}'$ and $\tilde{\sigma}''$ denote the generators of $H^*(G)$ and $H^*(G')$.

Proof. Strategy. Both sides live in $H^*(G \times G')$, where $(h_n \times h_{n'})^*$ is injective; so it is enough to compare their images in a polynomial ring. There the identity becomes the classical decomposition of an elementary symmetric function in $n + n'$ variables into products of elementary symmetric functions in the two groups of variables.

Step 1. A homotopy commutative square. Let

$$\tilde{F}: (\mathbb{RP}^\infty)^n \times (\mathbb{RP}^\infty)^{n'} \longrightarrow (\mathbb{RP}^\infty)^{n+n'}$$

be the identity under the evident identification. Both composites in

$$\begin{array}{ccc} (\mathbb{RP}^\infty)^n \times (\mathbb{RP}^\infty)^{n'} & \xrightarrow{\tilde{F}} & (\mathbb{RP}^\infty)^{n+n'} \\ \downarrow h_n \times h_{n'} & & \downarrow h_{n+n'} \\ G \times G' & \xrightarrow{F} & G'' \end{array}$$

classify $\gamma^1 \oplus \cdots \oplus \gamma^1$ with $n + n'$ summands, hence are homotopic by Theorem 4.5. Therefore

$$(h_n \times h_{n'})^* \circ F^* = \tilde{F}^* \circ h_{n+n'}^*.$$

Step 2. Computing the right-hand side. By definition we have $h_{n+n'}^*(\tilde{\sigma}_i) = \sigma_i(a_1, \dots, a_{n+n'})$. Under $\tilde{F}^*$ the first n variables become the variables $a'_1, \dots, a'_n$ of the first factor, and the last n' become $a''_1, \dots, a''_{n'}$. Sorting a monomial $a_{j_1} \cdots a_{j_i}$ by how many of its indices are at most n , we obtain

$$\sigma_i(a', a'') = \sum_{s=0}^i \sigma_s(a') \sigma_{i-s}(a'').$$

Step 3. Computing the left-hand side. By Künneth, $(h_n \times h_{n'})^*(x \times y) = h_n^*(x) \times h_{n'}^*(y)$, so

$$(h_n \times h_{n'})^* \left(\sum_s \tilde{\sigma}'_s \times \tilde{\sigma}''_{i-s} \right) = \sum_s \sigma_s(a') \sigma_{i-s}(a'').$$

Step 4. Conclusion. The two expressions agree, and $(h_n \times h_{n'})^*$ is injective, being a tensor product of injective maps between free modules (Theorem 5.15 and Künneth). Hence $F^*(\tilde{\sigma}_i) = \sum_s \tilde{\sigma}'_s \times \tilde{\sigma}''_{i-s}$. $\square$

Theorem 6.2 (Whitney product formula). *For vector bundles E, E' over the same base B ,*

$$w_i(E \oplus E') = \sum_{s=0}^i w_s(E) \smile w_{i-s}(E').$$

Proof. Strategy. Write the classifying map of $E \oplus E'$ as the composite of F with $f_E \times f_{E'}$ and the diagonal, then feed in Lemma 6.1 and convert cross products into cup products.

Step 1. The classifying map of the sum. Let $d: B \rightarrow B \times B$ be the diagonal. The bundle $(f_E \times f_{E'})^*(p_1^* \gamma^n \oplus p_2^* \gamma^{n'})$ over $B \times B$ restricts along d to $E \oplus E'$, so $F \circ (f_E \times f_{E'}) \circ d$ is a classifying map for $E \oplus E'$.

Step 2. Applying the lemma.

$$w_i(E \oplus E') = d^*(f_E \times f_{E'})^* F^*(\tilde{\sigma}_i) = d^*(f_E \times f_{E'})^* \left(\sum_s \tilde{\sigma}'_s \times \tilde{\sigma}''_{i-s} \right).$$

Step 3. Naturality of the cross product. $(f_E \times f_{E'})^*(x \times y) = f_E^* x \times f_{E'}^* y$, so the expression becomes $d^*(\sum_s w_s(E) \times w_{i-s}(E'))$.

Step 4. Diagonal. Finally $d^*(x \times y) = x \smile y$, which gives the formula. $\square$

This completes the axiomatic circle begun in Lecture 2: the Stiefel–Whitney classes exist, satisfy the four axioms, and are the only classes that do.

6.2 Complex vector bundles

Guiding question. *What changes when the fibres are complex vector spaces? Two things, both to our advantage: orientations come for free, and the relevant Grassmannians have cells only in even dimensions, so the theory can be run over $\mathbb{Z}$ instead of $\mathbb{Z}_2$.*

Definition 6.3. A *complex vector bundle* of rank n over B is a map $\pi: E \rightarrow B$ in which every fibre carries the structure of a complex vector space and local trivializations $\pi^{-1}(U) \cong U \times \mathbb{C}^n$ can be chosen complex linear on each fibre. All the constructions of Lecture 1 — pull-back, Whitney sum, sub-bundles — carry over verbatim.

Remark 6.4. A complex bundle of rank n is in particular a real bundle of rank $2n$. Conversely a real $\mathbb{R}^{2n}$ -bundle E can be promoted to a complex bundle exactly when it admits a *complex structure*: a fibrewise linear $J: E \rightarrow E$ with $J(J(v)) = -v$, scalar multiplication being defined by $(a + ib)v = av + bJv$. For instance TS^2 admits one, rotation by a quarter turn, whereas the tangent bundles of most even-dimensional spheres do not.

Definition 6.5. A *Hermitian metric* on a complex bundle is a Euclidean metric on the underlying real bundle satisfying $|iv| = |v|$. It determines a Hermitian inner product $\langle \cdot, \cdot \rangle$, complex linear in the first variable and conjugate linear in the second, and hence a notion of orthogonality. If $L \subset E$ is a complex sub-bundle then $L^\perp$ is again complex and $L \oplus L^\perp \cong E$ as complex bundles, by the argument of Theorem 1.24 applied with complex scalars.

Definition 6.6. The *complex Grassmannian* $G_n(\mathbb{C}^m)$ is the space of complex n -planes in $\mathbb{C}^m$, topologized as the quotient of the complex Stiefel manifold $V_n(\mathbb{C}^m)$ of complex n -frames. It is a compact manifold of real dimension $2n(m-n)$, and $G_1(\mathbb{C}^m) = \mathbb{CP}^{m-1}$. The tautological bundle $\gamma_n \rightarrow G_n(\mathbb{C}^m)$, the limit $G_n(\mathbb{C}^\infty)$ and its tautological bundle are defined exactly as in the real case, and the classification theorem holds with the same proof: complex n -bundles over a paracompact base B correspond to homotopy classes $[B, G_n(\mathbb{C}^\infty)]$.

Theorem 6.7. $G_n(\mathbb{C}^m)$ is a CW complex with one cell of real dimension $2 \sum_i (\sigma_i - i)$ for each Schubert symbol σ . All cells are even-dimensional, so the cellular boundary operator vanishes and

$$H^{2r}(G_n(\mathbb{C}^\infty); \mathbb{Z}) \cong \mathbb{Z}^{p_n(r)}, \quad H^{\text{odd}}(G_n(\mathbb{C}^\infty); \mathbb{Z}) = 0,$$

where $p_n(r)$ is the number of partitions of r into at most n parts.

Proof. Strategy. Repeat the Schubert analysis over $\mathbb{C}$. The only change is in

the normalization of a basis vector: a complex line contains a whole circle of unit vectors rather than two, so requiring the last non-zero coordinate to be real and positive cuts the relevant sphere down by one further dimension. The rotation argument of Lecture 5 goes through unchanged, and the parity of the resulting dimensions makes the chain complex collapse.

Step 1. Symbols. For $X \in G_n(\mathbb{C}^m)$ the sequence $\dim_{\mathbb{C}}(X \cap \mathbb{C}^k)$ increases from 0 to n in steps of at most 1, by the same exactness argument as in Claim 4.8, so it determines a symbol $1 \leq \sigma_1 < \dots < \sigma_n \leq m$ and a partition of $G_n(\mathbb{C}^m)$ into sets $e(\sigma)$.

Step 2. Normalized bases. Put

$$H_{\mathbb{C}}^{\sigma_i} = \{(x_1, \dots, x_{\sigma_i}, 0, \dots, 0) \in \mathbb{C}^m \mid x_{\sigma_i} \in \mathbb{R}_{>0}\}.$$

Given x in a complex line, the vectors $e^{i\theta}x$ all have the same norm, so there is a unique unit vector on the line whose σ_i -th coordinate is real and positive. Repeating the induction of Claim 4.9 with complex orthogonality gives: $X \in e(\sigma)$ if and only if X has a unique unitary basis $(x_1, \dots, x_n)$ with $x_i \in H_{\mathbb{C}}^{\sigma_i}$.

Step 3. Cell dimensions. For $n = 1$ the closure of $H_{\mathbb{C}}^{\sigma_1} \cap S^{2\sigma_1-1}$ consists of the unit vectors of $\mathbb{C}^{\sigma_1}$ with last coordinate in $\mathbb{R}_{\geq 0}$; this is a closed cell of real dimension $2\sigma_1 - 2$, since the last coordinate contributes one real parameter instead of two. For the inductive step, the space of admissible new vectors is the set of unit vectors in $\bar{H}_{\mathbb{C}}^{\sigma_{n+1}}$ orthogonal to n fixed ones, a closed cell of real dimension $2(\sigma_{n+1} - n) - 2$; the map Ψ built from unitary rotations $T(u, v)$ as in Lemma 5.2 identifies the product with $\bar{e}'(\sigma_1, \dots, \sigma_{n+1})$. Adding up gives real dimension $2 \sum_i (\sigma_i - i)$.

Step 4. Boundaries. The argument of Theorem 5.4 shows that the boundary of a cell lies in cells of strictly smaller dimension, so we do obtain a CW structure.

Step 5. Cohomology. Since all cells have even dimension, the cellular chain complex has zero groups in odd degrees, hence all differentials vanish and H_{2r} is free on the $2r$ -cells. Counting cells as in Lecture 5 gives $p_n(r)$ of them, and universal coefficients gives the statement for cohomology. $\square$

6.3 Chern classes

By Künneth, $H^*((\mathbb{CP}^\infty)^n; \mathbb{Z}) = \mathbb{Z}[a_1, \dots, a_n]$ with a_i of degree 2.

Theorem 6.8. *Let $h_n: (\mathbb{CP}^\infty)^n \rightarrow G_n(\mathbb{C}^\infty)$ classify $\gamma_1 \oplus \dots \oplus \gamma_1$. Then h_n^* is injective with image the symmetric polynomials, so*

$$H^*(G_n(\mathbb{C}^\infty); \mathbb{Z}) = \mathbb{Z}[\tilde{\sigma}_1, \dots, \tilde{\sigma}_n], \quad \tilde{\sigma}_i \in H^{2i}.$$

Proof. The proof of Theorem 5.15 applies word for word, with $\mathbb{RP}^\infty$ replaced by $\mathbb{CP}^\infty$, degrees doubled, and $\mathbb{Z}_2$ replaced by $\mathbb{Z}$. The three ingredients are

available: the splitting principle over $\mathbb{C}$ (Remark 6.12 below), the invariance of the image under permutation of the factors, and the cell count of Theorem 6.7, which here is an equality without any imported input, since the chain complex has no odd cells and hence zero differentials. $\square$

Definition 6.9 (Chern classes). For a complex bundle $E \rightarrow B$ of rank n with classifying map $f_E: B \rightarrow G_n(\mathbb{C}^\infty)$, set

$$c_i(E) := f_E^*(\tilde{\sigma}_i) \in H^{2i}(B; \mathbb{Z}), \quad c_0 = 1, \quad c_i = 0 \text{ for } i > n.$$

The *total Chern class* is $c(E) = 1 + c_1(E) + \cdots + c_n(E)$.

Theorem 6.10. *The Chern classes are the unique family of classes satisfying:*

- (C1) $c_0(E) = 1$ and $c_i(E) = 0$ for $i > \text{rk}_{\mathbb{C}} E$;
- (C2) $c_i(f^*E) = f^*c_i(E)$;
- (C3) $c_k(E_1 \oplus E_2) = \sum_j c_j(E_1) \smile c_{k-j}(E_2)$;
- (C4) $c_1(\gamma_1 \rightarrow \mathbb{CP}^1)$ is a generator of $H^2(\mathbb{CP}^1; \mathbb{Z})$.

Proof. Every step of Lectures 5 and 6 transposes: (C1) and (C2) hold by definition and by homotopy invariance of classifying maps; (C3) is proved exactly as Theorem 6.2, using the complex analogue of Lemma 6.1, whose proof is the same decomposition of elementary symmetric functions; and (C4) holds because $\tilde{\sigma}_1$ restricts to the generator on $G_1(\mathbb{C}^\infty) = \mathbb{CP}^\infty$ and then to $\mathbb{CP}^1$. Uniqueness follows as before: naturality reduces to the universal bundle, the complex splitting principle to sums of line bundles, (C3) to a single line bundle, and (C4) pins that case down. $\square$

Lemma 6.11. *Let $E \rightarrow B$ be a complex bundle of rank n over a base with a finite trivializing cover, and let $\mathbb{P}(E) \rightarrow B$ be its complex projectivization, with fibres $\mathbb{CP}^{n-1}$. Then*

$$H^*(\mathbb{P}(E); \mathbb{Z}) \cong H^*(B; \mathbb{Z}) \otimes H^*(\mathbb{CP}^{n-1}; \mathbb{Z}),$$

and in particular $H^(B; \mathbb{Z}) \rightarrow H^*(\mathbb{P}(E); \mathbb{Z})$ is injective.*

Proof. Identical to Lemma 5.10. The tautological complex line bundle γ over $\mathbb{P}(E)$ restricts on each fibre to the canonical bundle of $\mathbb{CP}^{n-1}$, so its classifying map $f: \mathbb{P}(E) \rightarrow \mathbb{CP}^\infty$ restricts on the fibre to a map homotopic to the standard inclusion, which is an isomorphism on H^{2k} for $k \leq n-1$. Setting $s(a^k) = f^*(a^k)$ gives $i_F^* \circ s = \text{id}$, and Theorem 5.7 applies over $\Lambda = \mathbb{Z}$, since $H^*(\mathbb{CP}^{n-1}; \mathbb{Z})$ is free of finite rank. $\square$

Remark 6.12. Consequently every complex bundle admits a splitting map: on $\mathbb{P}(E)$ the tautological line bundle is a complex sub-bundle of p^*E , and a

Hermitian metric splits $p^*E \cong \gamma \oplus \gamma^\perp$ with $\gamma^\perp$ of rank $n - 1$; iterate as in Corollary 5.12. Formally one writes

$$c(E) = \prod_{i=1}^n (1 + x_i), \quad x_i = c_1(L_i),$$

and calls the x_i the *Chern roots* of E . Every identity between Chern classes may be verified on these roots.

In the next lecture we change viewpoint entirely and construct the Stiefel–Whitney classes a second time, intrinsically, out of Steenrod squares and the Thom isomorphism. That route also hands us the Euler class, the integral refinement of the top class.

Lecture 7

Guiding question. *Milnor and Stasheff define the Stiefel–Whitney classes by the formula $w_i(\xi) = \phi^{-1} \text{Sq}^i \phi(1)$. What are the two ingredients Sq^i and ϕ , why does the formula satisfy our axioms, and what integral information appears along the way?*

7.1 Steenrod squares

Definition 7.1. The n -th homotopy group of a space X is $\pi_n(X) = [S^n, X]$. An *Eilenberg–MacLane space* $K(G, n)$ is a space with $\pi_n = G$ and all other homotopy groups trivial.

Theorem 7.2 (representability, quoted). *There is a natural bijection*

$$B: \langle X, K(G, n) \rangle \longrightarrow H^n(X; G), \quad [f] \longmapsto f^* \alpha,$$

where $\langle \ , \ \rangle$ denotes homotopy classes of maps and $\alpha \in H^n(K(G, n); G)$ is a fixed fundamental class.

We sketch the construction of the squares, following the course. Fix $\alpha \in H^n(X; \mathbb{Z}_2)$ and let $f_\alpha: X \rightarrow K(\mathbb{Z}_2, n)$ represent it. The cross square $\alpha \times \alpha \in H^{2n}(X \times X; \mathbb{Z}_2)$ is represented by a map $f_{\alpha \times \alpha}$, and the twist $T(x, y) = (y, x)$ satisfies $T^*(\alpha \times \alpha) = \alpha \times \alpha$ over $\mathbb{Z}_2$, so

$$f_{\alpha \times \alpha} \circ T \simeq f_{\alpha \times \alpha}.$$

Concatenating such a homotopy with its reverse gives a map $\tilde{F}: X \times X \times S^1 \rightarrow K(\mathbb{Z}_2, 2n)$ which is null-homotopic on $X \times X \times \{\text{pt}\}$; extending over D^2 , composing with T and gluing produces maps on $X \times X \times S^2$, and iterating yields

$$H: X \times X \times S^\infty \longrightarrow K(\mathbb{Z}_2, 2n), \quad H(x, y, s) = H(y, x, -s).$$

Restricting to the diagonal $x = y$, the symmetry makes H factor through $X \times \mathbb{RP}^\infty$, so we obtain a class in $H^{2n}(X \times \mathbb{RP}^\infty; \mathbb{Z}_2)$ which we expand using $H^*(\mathbb{RP}^\infty; \mathbb{Z}_2) = \mathbb{Z}_2[\omega]$:

$$H_\Delta^*(\alpha) = \sum_{i=0}^{2n} \alpha_i \times \omega^{2n-i}, \quad \alpha_i \in H^i(X; \mathbb{Z}_2).$$

Definition 7.3. $\text{Sq}^i(\alpha) := \alpha_{n+i} \in H^{n+i}(X; \mathbb{Z}_2)$.

Theorem 7.4 (properties, quoted). *The operations*

$$\text{Sq}^i: H^n(X; \mathbb{Z}_2) \longrightarrow H^{n+i}(X; \mathbb{Z}_2)$$

are natural homomorphisms satisfying

1. $\text{Sq}^i(f^*\alpha) = f^*(\text{Sq}^i\alpha)$ and $\text{Sq}^i(\alpha + \beta) = \text{Sq}^i\alpha + \text{Sq}^i\beta$;
2. $\text{Sq}^0 = \text{id}$; for $\alpha \in H^n$, $\text{Sq}^n\alpha = \alpha \smile \alpha$ and $\text{Sq}^i\alpha = 0$ for $i > n$;
3. the Cartan formula $\text{Sq}^i(\alpha \smile \beta) = \sum_j \text{Sq}^j\alpha \smile \text{Sq}^{i-j}\beta$.

The verification of these properties from the construction above is a long homotopy-theoretic argument which we do not reproduce; see Milnor–Stasheff §8 or Steenrod–Epstein.

7.2 Orientations

Definition 7.5. An *orientation* of a real n -dimensional vector space V is an equivalence class of bases, two bases being equivalent when the transition matrix has positive determinant. There are exactly two such classes.

A linearly embedded simplex $\Delta^n \subset V$ with ordered vertices $v_0, \dots, v_n$ determines the orientation given by $\overrightarrow{v_0v_1}, \dots, \overrightarrow{v_{n-1}v_n}$. Writing $V_0 = V \setminus \{0\}$ and assuming 0 lies in the interior of the simplex, the map $\sigma: \Delta^n \rightarrow V$ represents a generator of $H_n(V, V_0) \cong \mathbb{Z}$: this is the *preferred generator* μ_V attached to the orientation. Dually there is a preferred $u_V \in H^n(V, V_0)$ with $\langle u_V, \mu_V \rangle = 1$.

Definition 7.6. An *orientation of a vector bundle* ξ is a compatible choice of orientation on every fibre: local trivializations can be chosen orientation preserving on each fibre, equivalently all transition matrices have positive determinant. It determines a preferred generator $u_F \in H^n(F, F_0)$ on every fibre pair.

7.3 The Thom isomorphism

Theorem 7.7 (Thom isomorphism theorem). *Let $\pi: E \rightarrow B$ be a vector bundle of rank n , oriented if we use $\mathbb{Z}$ coefficients; over $\mathbb{Z}_2$ no hypothesis is needed. Write $E_0 = E \setminus (\text{zero section})$. Then*

1. *there is a unique class $u \in H^n(E, E_0)$, the Thom class, whose restriction to every fibre pair (F, F_0) is the preferred generator;*

2. $H^i(E, E_0) = 0$ for $i < n$, and

$$\phi: H^i(B) \xrightarrow{\pi^*} H^i(E) \xrightarrow{\smile^u} H^{i+n}(E, E_0)$$

is an isomorphism for every i .

The proof occupies the second half of Lecture 8. Here we record the formal consequences, which are what the applications actually use.

Theorem 7.8. *Let u, ϕ be as above.*

1. Pull-back. If $f: \xi \rightarrow \xi'$ is an orientation preserving bundle map then $f^*u' = u$.
2. Naturality. $\phi_\xi \circ \bar{f}^* = f^* \circ \phi_{\xi'}$.
3. Cross product. $u_{\xi \times \xi'} = u_\xi \times u_{\xi'}$ and $\phi_{\xi \times \xi'}(a \times b) = \phi_\xi(a) \times \phi_{\xi'}(b)$.

Proof. Strategy. All three are consequences of the uniqueness clause in Theorem 7.7: to identify a class in $H^n(E, E_0)$ it is enough to check what it restricts to on one fibre pair.

Step 1. Pull-back. The map f induces $f^*: H^n(E', E'_0) \rightarrow H^n(E, E_0)$. On a fibre pair, f restricts to an orientation preserving isomorphism $(F, F_0) \rightarrow (F', F'_0)$, which carries the preferred generator to the preferred generator. Hence f^*u' restricts to the preferred generator on every fibre of ξ , and uniqueness gives $f^*u' = u$.

Step 2. Naturality. For $y \in H^*(B')$,

$$f^*\phi_{\xi'}(y) = f^*(\pi'^*y \smile u') = \pi^*\bar{f}^*y \smile f^*u' = \pi^*\bar{f}^*y \smile u = \phi_\xi(\bar{f}^*y),$$

using that f^* is a ring map compatible with the relative cup product, $\pi' \circ f = \bar{f} \circ \pi$, and Step 1.

Step 3. Cross product. The class $u_\xi \times u_{\xi'}$ lies in $H^{n+m}(E \times E', (E \times E')_0)$, where $(E \times E')_0 = E_0 \times E' \cup E \times E'_0$. Restrict it to a fibre pair $(F^{n+m}, F_0^{n+m}) \cong (F^n \times F'^m, \dots)$: by the definition of the cross product and the product orientation,

$$(u_\xi \times u_{\xi'}) (\Delta^n \times \Delta^m) = u_\xi(\Delta^n) \cdot u_{\xi'}(\Delta^m) = 1,$$

so it is the preferred generator there. Uniqueness gives $u_{\xi \times \xi'} = u_\xi \times u_{\xi'}$, and the formula for ϕ follows by expanding both sides. $\square$

Definition 7.9 (Stiefel–Whitney classes, intrinsic definition). For an $\mathbb{R}^n$ -bundle ξ with $\mathbb{Z}_2$ Thom class $u = \phi(1)$, put

$$w_i(\xi) := \phi^{-1}(\text{Sq}^i u).$$

Proposition 7.10. *These classes satisfy the axioms of Definition 2.5, hence agree with the classes constructed in Lecture 5.*

Proof. Strategy. Match each axiom with the corresponding property of Sq from Theorem 7.4 and of ϕ from Theorem 7.8; then invoke the uniqueness theorem of Lecture 5.

Step 1. Rank. $\text{Sq}^0 u = u$ gives $w_0 = 1$. Since u has degree n , property (2) gives $\text{Sq}^i u = 0$ for $i > n$, so $w_i = 0$ for $i > n$.

Step 2. Naturality. For a bundle map f with base map $\bar{f}$, using naturality of Sq and Theorem 7.8,

$$w_i(\xi) = \phi_\xi^{-1} \text{Sq}^i u = \phi_\xi^{-1} \text{Sq}^i f^* u' = \phi_\xi^{-1} f^* \text{Sq}^i u' = \bar{f}^* \phi_{\xi'}^{-1} \text{Sq}^i u' = \bar{f}^* w_i(\xi').$$

Step 3. Whitney formula. By Theorem 7.8(3), $u_{\xi \times \xi'} = u_\xi \times u_{\xi'}$, and the Cartan formula applied to a cross product gives $\text{Sq}^i(u_\xi \times u_{\xi'}) = \sum_j \text{Sq}^j u_\xi \times \text{Sq}^{i-j} u_{\xi'}$. Applying $\phi_{\xi \times \xi'}^{-1}$ and using the product formula for ϕ yields $w_i(\xi \times \xi') = \sum_j w_j(\xi) \times w_{i-j}(\xi')$; pulling back along the diagonal converts this into the Whitney formula for $\xi \oplus \xi'$.

Step 4. Normalization. For $\gamma_1^1 \rightarrow \mathbb{RP}^1$ the pair (E, E_0) is the Möbius band relative to its boundary; there $\text{Sq}^1 u = u \smile u \neq 0$, as computed in Lecture 8, so $w_1(\gamma_1^1) \neq 0$.

Step 5. By the uniqueness theorem of Lecture 5, these classes coincide with $f_\xi^* \tilde{\sigma}_i$. □

7.4 The Euler class

Guiding question. *The Thom class lives in the relative group $H^n(E, E_0)$. What invariant of the base do we obtain by simply forgetting that it is relative?*

Definition 7.11. Let ξ be an oriented $\mathbb{R}^n$ -bundle over B with Thom class u . The *Euler class* is

$$e(\xi) := (\pi^*)^{-1} i^* u \in H^n(B; \mathbb{Z}),$$

where $i^*: H^n(E, E_0) \rightarrow H^n(E)$ forgets the pair and $\pi^*: H^*(B) \rightarrow H^*(E)$ is an isomorphism because π is a homotopy equivalence.

Theorem 7.12. *The Euler class has the following properties.*

1. Naturality: $\bar{f}^* e(\xi') = e(\xi)$ for an orientation preserving bundle map $f: \xi \rightarrow \xi'$.
2. If ξ is trivial then $e(\xi) = 0$.

3. Reversing the orientation of ξ changes the sign of $e(\xi)$; consequently $2e(\xi) = 0$ when n is odd.
4. $e(\xi) \bmod 2 = w_n(\xi)$.
5. $e(\xi \times \xi') = e(\xi) \times e(\xi')$ and $e(\xi \oplus \xi') = e(\xi) \smile e(\xi')$.
6. If ξ has a nowhere zero section then $e(\xi) = 0$.

Proof. Step 1. Apply i^* to $f^*u' = u$ and use $\pi^* \circ \bar{f}^* = f^* \circ \pi'^*$.

Step 2. A trivial bundle admits a bundle map to the bundle $\text{pt} \times \mathbb{R}^n$ over a point, whose Euler class lies in $H^n(\text{pt}) = 0$ for $n > 0$; now use (1).

Step 3. The map $(x, v) \mapsto (x, -v)$ is a bundle map over the identity which reverses the orientation of every fibre when n is odd, hence carries u to $-u$ and e to $-e$. Naturality then gives $e(\xi) = -e(\xi)$.

Step 4. Over $\mathbb{Z}_2$ we have $i^*u = \phi^{-1}$ applied to $u \smile u = \text{Sq}^n u$ up to the identification ϕ , so

$$e(\xi) \bmod 2 = \phi^{-1}(u \smile u) = \phi^{-1} \text{Sq}^n \phi(1) = w_n(\xi).$$

Step 5. By Theorem 7.8(3), $u_{\xi \times \xi'} = u_\xi \times u_{\xi'}$; applying i^* , which is compatible with cross products, and then $(\pi^*)^{-1}$, gives $e(\xi \times \xi') = e(\xi) \times e(\xi')$. Since $\xi \oplus \xi' = \Delta^*(\xi \times \xi')$ and $\Delta^*(x \times y) = x \smile y$, the second formula follows from (1).

Step 6. A nowhere zero section spans a trivial line sub-bundle $\ell \subset \xi$; with a metric, $\xi \cong \ell \oplus \ell^\perp$, so by (5) and (2), $e(\xi) = e(\ell) \smile e(\ell^\perp) = 0$. $\square$

Corollary 7.13. *If $2e(\xi) \neq 0$ then ξ does not split off an oriented summand of odd rank; in particular it has no nowhere zero section.*

In Lecture 8 we compute Euler classes of the canonical bundles, obtain the universal identity $e(\gamma^n) = \tilde{\sigma}_n$, and then prove Theorem 7.7.

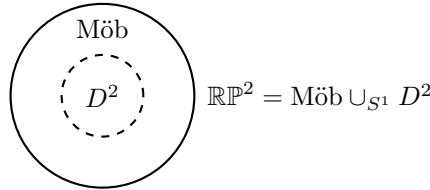
Lecture 8

Guiding question. *Two tasks. First, ground the Euler class in examples: canonical bundles over projective lines, and the universal identity $e(\gamma^n) = \tilde{\sigma}_n$. Second, pay the outstanding debt and prove the Thom isomorphism theorem.*

8.1 Euler classes of canonical bundles

Example 8.1 (The Möbius band). Consider $\mathbb{RP}^1 \hookrightarrow \mathbb{RP}^2$, $[x_0, x_1] \mapsto [x_0, x_1, 0]$. The complement of $\mathbb{RP}^1$ is the affine chart $\{x_2 \neq 0\} \cong \mathbb{R}^2$, an open disc, and a neighbourhood of $\mathbb{RP}^1$ in $\mathbb{RP}^2$ is the total space of its normal bundle, which is γ_1^1 . Hence

$$\mathbb{RP}^2 = (\text{Möbius band}) \cup_{\partial} D^2.$$



The pair (E, E_0) for γ_1^1 deformation retracts onto (Möbius band, boundary), so with $\mathbb{Z}_2$ coefficients

$$H^{*>0}(\gamma_1^1, (\gamma_1^1)_0) \cong H^{*>0}(\mathbb{RP}^2) = \mathbb{Z}_2[a]/(a^3),$$

by excising the complementary disc. Under this identification the Thom class u corresponds to a , so $u \smile u$ corresponds to $a^2 \neq 0$. This is the computation quoted in Step 4 of the previous lecture, and it gives $w_1(\gamma_1^1) \neq 0$ once more, now by Thom-theoretic means.

Example 8.2 (The complex canonical bundle). Let $\gamma_1 \rightarrow \mathbb{CP}^1$ be the tautological complex line bundle. Exactly as above, $\mathbb{CP}^1 \subset \mathbb{CP}^2$ has normal bundle γ_1 and complementary affine chart a 4-disc, so

$$\mathbb{CP}^2 = E(\gamma_1) \cup_{S^3} D^4,$$

and $H^*(E, E_0; \mathbb{Z}) \cong H^{*>0}(\mathbb{CP}^2; \mathbb{Z})$, generated by u in degree 2 with u^2 generating degree 4. Chasing u through $H^2(E, E_0) \rightarrow H^2(E) \cong H^2(\mathbb{CP}^1)$ identifies

$$e(\gamma_1 \rightarrow \mathbb{CP}^1) = a,$$

a generator of $H^2(\mathbb{CP}^1; \mathbb{Z})$. Note that every complex bundle is canonically oriented, by declaring $v_1, Jv_1, \dots, v_n, Jv_n$ positive for any complex basis $v_1, \dots, v_n$; so the Euler class of a complex bundle is defined without extra choices.

Theorem 8.3. $e(\gamma_n) = \tilde{\sigma}_n = c_n(\gamma_n)$ for the universal complex bundle, and over $\mathbb{Z}_2$, $e(\gamma^n) = \tilde{\sigma}_n = w_n(\gamma^n)$ for the universal real bundle. Consequently, for every bundle ξ ,

$$e(\xi) = c_n(\xi) \quad (\xi \text{ complex}), \quad e(\xi) \bmod 2 = w_n(\xi) \quad (\xi \text{ real oriented}).$$

Proof. Strategy. Compute the Euler class of a sum of n line bundles over $(\mathbb{CP}^\infty)^n$, where multiplicativity turns it into $a_1 \cdots a_n = \sigma_n$; then use that h_n^* is injective to transport the identity to the universal bundle.

Step 1. Stabilizing. The inclusion $\mathbb{CP}^1 \hookrightarrow \mathbb{CP}^\infty$ is covered by a bundle map $\gamma_1 \rightarrow \gamma_1$ and induces an isomorphism on H^2 . By naturality and Example 8.2, $e(\gamma_1 \rightarrow \mathbb{CP}^\infty) = a$, the generator of $H^2(\mathbb{CP}^\infty; \mathbb{Z}) = \mathbb{Z}[a]$.

Step 2. Sums of lines. Over $(\mathbb{CP}^\infty)^n$, by Theorem 7.12(5),

$$e(\gamma_1 \oplus \cdots \oplus \gamma_1) = e(\gamma_1) \smile \cdots \smile e(\gamma_1) = a_1 a_2 \cdots a_n = \sigma_n(a_1, \dots, a_n).$$

Step 3. Transport. Let $h_n: (\mathbb{CP}^\infty)^n \rightarrow G_n(\mathbb{C}^\infty)$ classify that sum, so $h_n^* \gamma_n \cong \gamma_1 \oplus \cdots \oplus \gamma_1$. By naturality of the Euler class, $h_n^* e(\gamma_n) = \sigma_n = h_n^*(\tilde{\sigma}_n)$, and h_n^* is injective by Theorem 6.8; hence $e(\gamma_n) = \tilde{\sigma}_n = c_n(\gamma_n)$.

Step 4. Real case and general bundles. The same three steps over $\mathbb{RP}^\infty$ with $\mathbb{Z}_2$ coefficients give $e(\gamma^n) = \tilde{\sigma}_n$. For an arbitrary bundle, pull back along a classifying map and use naturality of both e and the characteristic classes. $\square$

8.2 Proof of the Thom isomorphism theorem

We now prove Theorem 7.7. Throughout, $\pi: E \rightarrow B$ has rank n and E_0 is the complement of the zero section. The model computation is the following consequence of Künneth.

Lemma 8.4. Let $e^1 \in H^1(\mathbb{R}, \mathbb{R}_0; \Lambda)$ be a generator, where $\mathbb{R}_0 = \mathbb{R} \setminus 0$ and Λ is any coefficient ring. Then for every space B and every open $B' \subset B$ the map

$$H^j(B, B') \longrightarrow H^{j+1}(B \times \mathbb{R}, B' \times \mathbb{R} \cup B \times \mathbb{R}_0), \quad y \longmapsto y \times e^1,$$

is an isomorphism; iterating n times, with $e^n = e^1 \times \cdots \times e^1$,

$$H^j(B, B') \xrightarrow{\cong} H^{j+n}(B \times \mathbb{R}^n, B' \times \mathbb{R}^n \cup B \times \mathbb{R}_0^n), \quad y \longmapsto y \times e^n.$$

Proof. The pair $(\mathbb{R}, \mathbb{R}_0)$ has cohomology Λ concentrated in degree 1, generated by e^1 , and it is free; the relative Künneth theorem therefore gives

$$H^{j+1}(B \times \mathbb{R}, B' \times \mathbb{R} \cup B \times \mathbb{R}_0) \cong \bigoplus_{p+q=j+1} H^p(B, B') \otimes H^q(\mathbb{R}, \mathbb{R}_0) = H^j(B, B') \otimes \Lambda,$$

and the isomorphism is exactly $y \mapsto y \times e^1$. Iterate. $\square$

Proof of Theorem 7.7. Strategy. Four stages of increasing generality. For a trivial bundle the statement is Lemma 8.4. If the base splits into two open pieces over which the theorem is known, Mayer–Vietoris and the five lemma give it over the union; a finite trivializing cover then handles compact bases. Arbitrary bases follow by writing cohomology as a limit over compact subsets, which works with field coefficients. Finally an algebraic lemma about mapping cones upgrades field coefficients to $\mathbb{Z}$.

Step 1. Case 1: E trivial. Here $(E, E_0) \cong (B \times \mathbb{R}^n, B \times \mathbb{R}_0^n)$. Taking $B' = \emptyset$ in Lemma 8.4 gives

$$H^j(B) \cong H^{j+n}(E, E_0), \quad y \mapsto y \times e^n.$$

In particular $H^{j+n}(E, E_0) = 0$ for $j < 0$, and $H^n(E, E_0) \cong H^0(B)$. Put $u = 1 \times e^n$. Its restriction to a fibre pair is e^n , the preferred generator, so u is a Thom class; and it is unique because a class restricting to 0 on every fibre corresponds to $y \times e^n$ with y vanishing at every point of B , hence $y = 0$. Finally, elements of $H^j(E) = H^j(B \times \mathbb{R}^n)$ are of the form $y \times 1$, and

$$(y \times 1) \smile u = (y \times 1) \smile (1 \times e^n) = y \times e^n,$$

so $\smile u$ is the isomorphism just described.

Step 2. Case 2: Mayer–Vietoris. Suppose $B = B' \cup B''$ with B', B'' open and the theorem known over B', B'' and $B^\cap = B' \cap B''$. Write $E', E'', E^\cap$ for the restrictions of E . Consider the Mayer–Vietoris sequence of the pairs,

$$\begin{aligned} H^{n-1}(E^\cap, E_0^\cap) &\longrightarrow H^n(E, E_0) \xrightarrow{(i'^*, i''^*)} H^n(E', E_0') \oplus H^n(E'', E_0'') \\ &\xrightarrow{i'^* - i''^*} H^n(E^\cap, E_0^\cap). \end{aligned}$$

By hypothesis $H^{n-1}(E^\cap, E_0^\cap) \cong H^{-1}(B^\cap) = 0$. The classes u', u'' both restrict to the unique Thom class $u^\cap$ of $E^\cap$, so the pair (u', u'') dies under $i'^* - i''^*$ and lifts to a unique $u \in H^n(E, E_0)$; it restricts to the preferred generator on every fibre, since every fibre lies over B' or B'' . Now $\smile u$ maps the Mayer–Vietoris sequence of (B', B'') to that of (E', E'') , commutatively; the maps over $B', B'', B^\cap$ are isomorphisms by hypothesis, so the five lemma gives the result over B .

Step 3. Case 3: B compact. Choose a finite trivializing cover $U_1, \dots, U_r$. By Case 1 the theorem holds over each U_i and over each intersection, which is

again trivializing. By Case 2 it holds over $U_1 \cup U_2$; inductively, if it holds over $U_1 \cup \dots \cup U_j$ then apply Case 2 to that set together with U_{j+1} , whose intersection with it is a union of j trivializing sets covered by the inductive hypothesis. After r steps we reach B .

Step 4. Case 4: general base, field coefficients. Every singular chain has compact image, so $H_j(B) = \varinjlim_C H_j(C)$ over the compact subsets $C \subset B$. With field coefficients Λ , cohomology is the dual of homology, and duals turn direct limits into inverse limits:

$$H^j(B; \Lambda) = \text{Hom} \left(\varinjlim_C H_j(C), \Lambda \right) = \varprojlim_C \text{Hom} \left(H_j(C), \Lambda \right) = \varprojlim_C H^j(C; \Lambda),$$

and similarly for the pairs (E, E_0) restricted over C . By Case 3 there is a compatible Thom class over each C , and compatible isomorphisms; passing to the inverse limit gives a Thom class u over B , its uniqueness, and the isomorphism $\smile u$. This already proves the theorem over $\mathbb{Z}_2$ for arbitrary bundles, which is all that Stiefel–Whitney theory requires.

Step 5. Integral coefficients. Two further ingredients are needed, and they occupy the rest of the lecture: the algebraic Lemma 8.5 below, and the vanishing $H_{n-1}(E, E_0; \mathbb{Z}) = 0$. Granting the latter, universal coefficients,

$$0 \rightarrow \text{Ext} \left(H_{n-1}(E, E_0), \mathbb{Z} \right) \rightarrow H^n(E, E_0; \mathbb{Z}) \rightarrow \text{Hom} \left(H_n(E, E_0), \mathbb{Z} \right) \rightarrow 0,$$

gives $H^n(E, E_0; \mathbb{Z}) \cong \text{Hom}(H_n(E, E_0), \mathbb{Z})$, and the compatible fibrewise normalization determines a unique integral u . Choose a singular cocycle $z \in Z^n(E, E_0)$ representing u . Cap product with z is a chain map of degree $-n$,

$$z \cap : C_{n+i}(E, E_0) \longrightarrow C_i(E), \quad \partial(z \cap \gamma) = (-1)^n z \cap \partial \gamma,$$

whose dual on cochains is $c \mapsto c \smile z$. By Step 4 the induced map on cohomology is an isomorphism over every field. By Lemma 8.5 it is then an isomorphism over $\mathbb{Z}$, and over any coefficient ring. $\square$

Lemma 8.5. *Let $f: K \rightarrow K'$ be a chain map of free chain complexes over $\mathbb{Z}$ which induces isomorphisms $H^*(K'; \Lambda) \rightarrow H^*(K; \Lambda)$ for every field Λ . Then f induces isomorphisms on homology and cohomology with any coefficient ring.*

Proof. Strategy. Encode “ f is a homology isomorphism” as the vanishing of the homology of the algebraic mapping cone. That homology is finitely constrained by universal coefficients: rational coefficients show it is torsion, and $\mathbb{Z}_p$ coefficients show it has no p -torsion for any prime p .

Step 1. The mapping cone. Let d be the degree of f and put $K_i^f = K_{i-d-1} \oplus K'_i$ with

$$\partial^f(k, k') = ((-1)^{d+1} \partial k, f(k) + \partial k').$$

A direct computation gives $\partial^f \circ \partial^f = 0$, using $\partial f = (-1)^d f \partial$. The complex K^f is free.

Step 2. The long exact sequence. There is a short exact sequence of complexes $0 \rightarrow K' \rightarrow K^f \rightarrow K[d+1] \rightarrow 0$, whose long exact sequence has connecting homomorphism induced by f :

$$\cdots \rightarrow H_i(K') \rightarrow H_i(K^f) \rightarrow H_{i-d-1}(K) \xrightarrow{f_*} H_{i-1}(K') \rightarrow \cdots$$

Hence $H_*(K^f) = 0$ if and only if f_* is an isomorphism on homology.

Step 3. Vanishing over every field. By hypothesis f induces isomorphisms on cohomology with field coefficients, so the corresponding sequence gives $H^*(K^f; \Lambda) = 0$ for every field Λ . Universal coefficients,

$$0 \rightarrow \text{Ext}(H_{q-1}(K^f), \Lambda) \rightarrow H^q(K^f; \Lambda) \rightarrow \text{Hom}(H_q(K^f), \Lambda) \rightarrow 0,$$

then forces $\text{Hom}(H_q(K^f), \Lambda) = 0$ for every field.

Step 4. Conclusion. Taking $\Lambda = \mathbb{Q}$ shows $H_q(K^f)$ has no free part, so it is a torsion group. Taking $\Lambda = \mathbb{Z}_p$ and using also the Ext term shows $H_q(K^f)$ has no p -torsion, for every prime p . Hence $H_q(K^f) = 0$ for all q , and by Step 2 the map f is a homology isomorphism over $\mathbb{Z}$; the same then holds with any coefficients, again by universal coefficients applied to the free complexes. $\square$

Corollary 8.6. *For an oriented rank n bundle, the cap product with the Thom class,*

$$H_{n+i}(E, E_0; \Lambda) \longrightarrow H_i(E; \Lambda), \quad \eta \longmapsto u \cap \eta,$$

is an isomorphism for every coefficient ring Λ , compatibly with the cohomological Thom isomorphism through $\langle c, u \cap \gamma \rangle = \langle c \smile u, \gamma \rangle$.

Remark 8.7. It remains to justify $H_{n-1}(E, E_0; \mathbb{Z}) = 0$, used in Step 5. For compact B this follows from Corollary 8.6 in the already established cases, which gives $H_{n-1}(E, E_0) \cong H_{-1}(E) = 0$; for general B write $H_{n-1}(E, E_0; \mathbb{Z}) = \varinjlim_C H_{n-1}(E|_C, E_0|_C; \mathbb{Z}) = 0$, since homology commutes with direct limits.

With the Thom isomorphism established, the next lecture brings it into contact with geometry: tubular neighbourhoods, dual classes of submanifolds, and the diagonal.

Lecture 9

Guiding question. *The Thom class lives on the total space of a bundle. Geometry enters through one observation: a submanifold has a neighbourhood that looks exactly like its normal bundle. This transports the Thom class into the cohomology of the ambient manifold, where it becomes the dual class of the submanifold. Applied to the diagonal in $M \times M$ it will compute the Euler class of TM .*

Throughout, $M^n \subset A^{n+k}$ is a closed smooth n -manifold embedded in a smooth Riemannian $(n+k)$ -manifold, and

$$\nu = \{v \in T_x A \mid x \in M, v \perp T_x M\}$$

is its normal bundle, so that $TA|_M \cong TM \oplus \nu$.

9.1 The tubular neighbourhood theorem

Theorem 9.1. *There is an open neighbourhood of M in A diffeomorphic to the total space $E(\nu)$, by a diffeomorphism restricting to the identity on the zero section.*

Proof. Strategy. Write down a map from the normal bundle to A which is the identity on M and whose differential along M is invertible, then quote the global inverse function theorem. The map is the exponential map of the Riemannian metric.

Step 1. The inverse function theorem, global form (quoted). If $f: N \rightarrow A$ is smooth, injective on a compact submanifold $M \subset N$, and df_x is an isomorphism for every $x \in M$, then f maps some neighbourhood of M in N diffeomorphically onto a neighbourhood of $f(M)$ in A .

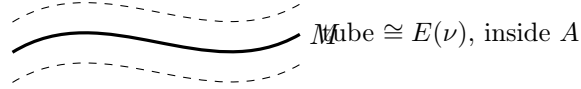
Step 2. The exponential map. For $\varepsilon > 0$ let $E(\varepsilon) = \{v \in E(\nu) \mid |v| < \varepsilon\}$, an open neighbourhood of the zero section, and define

$$\text{Exp}: E(\varepsilon) \longrightarrow A, \quad (x, v) \longmapsto \gamma_v(1),$$

where γ_v is the unique geodesic with $\gamma_v(0) = x$ and $\dot{\gamma}_v(0) = v$. For ε small this is defined and smooth, by the theory of ordinary differential equations.

Step 3. Behaviour along M . On the zero section Exp is the identity. Its differential at a point of the zero section, under the canonical splitting $T(E(\nu))|_M \cong TM \oplus \nu \cong TA|_M$, is the identity: the TM directions are unmoved and the ν directions differentiate the geodesics, giving back v .

Step 4. Conclusion. Step 1 applies with $N = E(\varepsilon)$ and $f = \text{Exp}$, so a neighbourhood of the zero section in $E(\varepsilon)$ maps diffeomorphically onto a neighbourhood U of M in A . Finally $E(\nu) \cong E(\varepsilon)$ by the fibrewise rescaling $v \mapsto \varepsilon v / \sqrt{1 + |v|^2}$. $\square$



Corollary 9.2. *If M is closed in A then, for any coefficient ring,*

$$H^*(E, E_0) \cong H^*(A, A \setminus M), \quad E = E(\nu).$$

Proof. Let $U \subset A$ be a tubular neighbourhood, so $(U, U \setminus M) \cong (E, E_0)$ by Theorem 9.1. Excising the closed set $A \setminus U$ from the pair $(A, A \setminus M)$ gives $H^*(A, A \setminus M) \cong H^*(U, U \setminus M)$. $\square$

9.2 The dual class of a submanifold

Assume from now on either $\mathbb{Z}_2$ coefficients, or that TA and TM are oriented and coefficients are $\mathbb{Z}$; in the latter case ν inherits an orientation from $TA|_M \cong TM \oplus \nu$. By Theorem 7.7 there is a unique Thom class $u \in H^k(E, E_0)$, and Corollary 9.2 carries it to a class

$$u' \in H^k(A, A \setminus M),$$

whose image in $H^k(A)$ we denote u'' . Both are called the *dual class* of M in A .

Remark 9.3. The class u' does not depend on the Riemannian metric: two metrics g_1, g_2 are joined by the path $tg_1 + (1 - t)g_2$ of metrics, and the corresponding exponential identifications are homotopic.

Theorem 9.4. *The inclusions $M \hookrightarrow A \hookrightarrow (A, A \setminus M)$ induce*

$$H^k(A, A \setminus M) \longrightarrow H^k(A) \longrightarrow H^k(M),$$

under which $u' \mapsto u''$ and u'' restricts to

$$\begin{cases} e(\nu) & \text{in the oriented case,} \\ w_k(\nu) & \text{over } \mathbb{Z}_2. \end{cases}$$

Proof. Restrict the whole picture to the tube. The composite $H^k(E, E_0) \rightarrow H^k(E) \cong H^k(M)$ sends u to $e(\nu)$, by the very definition of the Euler class; over $\mathbb{Z}_2$ it sends u to $w_k(\nu)$ by Theorem 7.12(4). The square

$$\begin{array}{ccccc} H^k(A, A \setminus M) & \longrightarrow & H^k(A) & \longrightarrow & H^k(M) \\ \cong \downarrow & & \downarrow & & \parallel \\ H^k(E, E_0) & \longrightarrow & H^k(E) & \xrightarrow{\cong} & H^k(M) \end{array}$$

commutes, since all maps are induced by inclusions and the tubular identification. $\square$

Corollary 9.5. *If M^n is closed and smoothly embedded in $\mathbb{R}^{n+k}$, then $w_k(\nu) = 0$, and in the oriented case $e(\nu) = 0$.*

Proof. Here u'' lies in $H^k(\mathbb{R}^{n+k}) = 0$ for $k > 0$, so its restriction to M vanishes; apply Theorem 9.4. $\square$

Example 9.6. Combined with Whitney duality this refines the immersion obstructions of Lecture 2 into *embedding* obstructions. Since $\bar{w}(T\mathbb{RP}^{2^r}) = 1 + a + \dots + a^{2^r-1}$ has non-zero term in degree $2^r - 1$, an embedding $\mathbb{RP}^{2^r} \subset \mathbb{R}^{2^r+k}$ would give $w_j(\nu) = \bar{w}_j(T\mathbb{RP}^{2^r}) = 0$ for $j > k$ and, by Corollary 9.5, also $w_k(\nu) = 0$; hence $k \geq 2^r$. So $\mathbb{RP}^{2^r}$ does not embed in $\mathbb{R}^{2 \cdot 2^r - 1}$, which is sharp against Whitney's embedding theorem $M^n \hookrightarrow \mathbb{R}^{2n}$.

9.3 The diagonal

Apply the machine to the diagonal embedding $\Delta: M \rightarrow M \times M$, $x \mapsto (x, x)$, of a closed Riemannian manifold; the product carries the product metric.

Lemma 9.7. *The normal bundle $N(\Delta)$ of $\Delta(M) \subset M \times M$ is canonically isomorphic to TM .*

Proof. At (x, x) the tangent space of the diagonal is $T\Delta = \{(v, v)\} \subset T_x M \times T_x M$, whose orthogonal complement is $N(\Delta) = \{(-v, v)\}$. The map

$$TM \longrightarrow N(\Delta), \quad (x, v) \longmapsto ((x, x), (-v, v)),$$

is continuous, fibrewise a linear isomorphism, hence an isomorphism of bundles by Lemma 1.10. $\square$

Corollary 9.8. *Let $u'' \in H^n(M \times M)$ be the dual class of the diagonal. Then*

$$\Delta^* u'' = e(TM) \quad (\text{resp. } w_n(TM) \text{ over } \mathbb{Z}_2).$$

Proof. Theorem 9.4 identifies the restriction of u'' to $\Delta(M)$ with $e(N(\Delta))$, and Lemma 9.7 identifies $N(\Delta)$ with TM . Restricting to $\Delta(M)$ is the same as applying Δ^* . $\square$

The dual class of the diagonal interacts with the product structure very rigidly.

Lemma 9.9. *For every $a \in H^*(M)$,*

$$(a \times 1) \smile u'' = (1 \times a) \smile u'' \in H^*(M \times M).$$

Proof. Strategy. Both products are computed from classes supported near the diagonal, and near the diagonal the two projections $M \times M \rightarrow M$ agree. Make this precise by factoring the cup product through the relative group and restricting to a tubular neighbourhood, which retracts onto the diagonal.

Step 1. Where the product is computed. Let U be a tubular neighbourhood of $\Delta(M)$ in $M \times M$. Cup product with the relative class u' gives a commutative square

$$\begin{array}{ccc} H^i(M \times M) & \longrightarrow & H^i(U) \\ \smile u' \downarrow & & \downarrow \smile u'|_U \\ H^{i+n}(M \times M, M \times M \setminus \Delta) & \xrightarrow[\cong]{\text{excision}} & H^{i+n}(U, U \setminus \Delta) \end{array}$$

so $x \smile u'$ depends only on the restriction $x|_U$.

Step 2. The two restrictions agree. The diagonal is a deformation retract of U , and $p_1 \circ \Delta = p_2 \circ \Delta = \text{id}$; hence the two maps $p_1, p_2: U \rightarrow M$ are homotopic, and for $a \in H^*(M)$,

$$(a \times 1)|_U = p_1^* a|_U = p_2^* a|_U = (1 \times a)|_U.$$

Step 3. Conclusion. By Steps 1 and 2, $(a \times 1) \smile u' = (1 \times a) \smile u'$ in $H^*(M \times M, M \times M \setminus \Delta)$. Mapping to the absolute group $H^*(M \times M)$ turns u' into u'' and preserves the identity. $\square$

9.4 The slant product

We work with field coefficients, so that $H^*(X \times Y) = H^*(X) \otimes H^*(Y)$.

Definition 9.10. The *slant product*

$$/ : H^{p+q}(X \times Y) \otimes H_q(Y) \longrightarrow H^p(X)$$

is defined on decomposable classes by $(\alpha \times \beta)/\mu = \langle \beta, \mu \rangle \alpha$ and extended linearly.

Lemma 9.11. For $a \in H^*(X)$, $P \in H^*(X \times Y)$ and $\mu \in H_*(Y)$,

$$((a \times 1) \smile P) / \mu = a \smile (P / \mu).$$

Proof. Write $P = \sum_{i,j} a_{ij} p_i \times q_j$ with $p_i \in H^*(X)$, $q_j \in H^*(Y)$. Then

$$(a \times 1) \smile P = \sum_{i,j} a_{ij} (a \smile p_i) \times q_j,$$

so slanting with μ gives $\sum_{i,j} a_{ij} \langle q_j, \mu \rangle (a \smile p_i) = a \smile (\sum_{i,j} a_{ij} \langle q_j, \mu \rangle p_i) = a \smile (P / \mu)$. $\square$

Lemma 9.12. Let $\mu \in H_n(M)$ be the fundamental class. Then

$$u'' / \mu = 1 \in H^0(M).$$

The proof is a local computation which opens the next lecture. Together with Lemma 9.9 and Corollary 9.8 it will determine u'' completely and produce the identity $\langle e(TM), \mu \rangle = \chi(M)$.

Lecture 10

Guiding question. *We finish the diagonal computation: the dual class of the diagonal is a sum over dual bases, and evaluating it turns $e(TM)$ into the Euler characteristic. Then we introduce the last family of characteristic classes, the Pontryagin classes of a real bundle.*

Throughout M is a closed n -manifold with fundamental class μ ; coefficients are a field, always admissible over $\mathbb{Z}_2$ and, when M is oriented, over any field.

10.1 The local computation

Proof of Lemma 9.12. Strategy. The class u''/μ lies in $H^0(M)$, so it is determined by its value at each point x . Naturality of the slant product turns that value into the evaluation of the dual class against the local fundamental class at x , and the fibre-restriction property of the Thom class makes it 1.

Step 1. Reduction to a point. Fix $x \in M$ and let $i_x: M \rightarrow M \times M$, $y \mapsto (x, y)$. Naturality of the slant product in the first variable gives that the value of u''/μ at x is

$$(i_x^* u'')/\mu = \langle i_x^* u'', \mu \rangle,$$

where we use that $i_x^* u'' \in H^n(M)$ and that slanting an n -class with the n -cycle μ is the Kronecker pairing.

Step 2. Passing to the relative class. Let $j_x: (M, M \setminus x) \rightarrow (M \times M, M \times M \setminus \Delta)$ be $y \mapsto (x, y)$; it is well defined, since $(x, y) \in \Delta$ forces $y = x$. Then $i_x^* u'' = i_M^* j_x^* u'$, where $i_M: M \rightarrow (M, M \setminus x)$, so with $\mu_x \in H_n(M, M \setminus x)$ the image of μ ,

$$\langle i_x^* u'', \mu \rangle = \langle j_x^* u', \mu_x \rangle.$$

Step 3. Identifying $j_x^ u'$.* Under the exponential chart of Theorem 9.1, the map j_x corresponds to the inclusion of the fibre of $N(\Delta)$ over (x, x) :

$$(N_x, N_x \setminus 0) \xrightarrow{\text{Exp}} (M, M \setminus x) \xrightarrow{j_x} (M \times M, M \times M \setminus \Delta),$$

the first map being a homeomorphism onto a neighbourhood, and the composite being homotopic to $v \mapsto (x, \exp_x v)$. Since u' comes from the Thom

class, its restriction to a fibre pair is the preferred generator; hence $j_x^* u'$ is the preferred generator of $H^n(T_x M, T_x M \setminus 0)$.

Step 4. Conclusion. The correspondence between orientations of $T_x M$ and local orientations of M is exactly the one induced by $\exp$, so μ_x corresponds to the preferred generator in homology and $\langle j_x^* u', \mu_x \rangle = 1$. Therefore u''/μ takes the value 1 at every x , that is, $u''/\mu = 1$. $\square$

10.2 The duality theorem and the Euler characteristic

Theorem 10.1 (Poincaré duality, quoted). *For M closed, and with field coefficients (any field if M is oriented, $\mathbb{Z}_2$ otherwise), the cup product pairing $H^i(M) \otimes H^{n-i}(M) \rightarrow \Lambda$, $(x, y) \mapsto \langle x \smile y, \mu \rangle$, is non-degenerate. Equivalently, every homogeneous basis $b_1, \dots, b_r$ of $H^*(M)$ admits a dual basis $b_1^\#, \dots, b_r^\#$ with*

$$\langle b_i \smile b_j^\#, \mu \rangle = \delta_{ij}.$$

Theorem 10.2. *In terms of such bases the dual class of the diagonal is*

$$u'' = \sum_{i=1}^r (-1)^{\dim b_i} b_i \times b_i^\#.$$

Proof. Strategy. Expand u'' in the Künneth basis with unknown coefficients, then exploit the two facts proved so far: u'' slants to 1 against μ , and multiplying by $a \times 1$ or $1 \times a$ gives the same answer. Evaluating on basis elements identifies the unknowns as the dual basis.

Step 1. Expansion. By Künneth, $H^n(M \times M) = \bigoplus_i H^{\dim b_i}(M) \otimes H^{n-\dim b_i}(M)$, so we may write

$$u'' = \sum_{i=1}^r b_i \times c_i, \quad \dim b_i + \dim c_i = n.$$

Step 2. A formula for every class. Let $a \in H^*(M)$. Using Lemma 9.11, then Lemma 9.9, then Lemma 9.12,

$$a = a \smile (u''/\mu) = ((a \times 1) \smile u'')/\mu = ((1 \times a) \smile u'')/\mu.$$

Step 3. Expanding the right-hand side. Substituting the expansion of u'' and moving a past b_i , which costs the Koszul sign $(-1)^{\dim a \dim b_i}$,

$$\begin{aligned} ((1 \times a) \smile u'')/\mu &= \sum_i (-1)^{\dim a \dim b_i} (b_i \times (a \smile c_i))/\mu \\ &= \sum_i (-1)^{\dim a \dim b_i} \langle a \smile c_i, \mu \rangle b_i. \end{aligned}$$

Step 4. Identification. Take $a = b_j$. Comparing with Step 2, which says the sum equals b_j , and using that the b_i are a basis,

$$(-1)^{\dim b_j \dim b_i} \langle b_j \smile c_i, \mu \rangle = \delta_{ij}.$$

For $i = j$ the sign is $(-1)^{(\dim b_i)^2} = (-1)^{\dim b_i}$, so $\langle b_i \smile c_i, \mu \rangle = (-1)^{\dim b_i}$, while for $i \neq j$ the pairing vanishes. Comparing with Theorem 10.1 gives $c_i = (-1)^{\dim b_i} b_i^\#$. $\square$

Theorem 10.3. *For M closed and oriented,*

$$\langle e(TM), \mu \rangle = \chi(M) = \sum_k (-1)^k \dim H^k(M);$$

without orientation, $\langle w_n(TM), \mu \rangle \equiv \chi(M) \pmod{2}$.

Proof. By Corollary 9.8 and Theorem 10.2,

$$e(TM) = \Delta^* u'' = \sum_i (-1)^{\dim b_i} \Delta^* (b_i \times b_i^\#) = \sum_i (-1)^{\dim b_i} b_i \smile b_i^\#.$$

Evaluating on μ and using $\langle b_i \smile b_i^\#, \mu \rangle = 1$,

$$\langle e(TM), \mu \rangle = \sum_i (-1)^{\dim b_i} = \sum_k (-1)^k \dim H^k(M) = \chi(M),$$

since the b_i of a given dimension k form a basis of $H^k(M)$. $\square$

Example 10.4. $\chi(S^2) = 2$, so $e(TS^2) \neq 0$: a third proof that TS^2 is non-trivial, and the statement underlying the hairy ball theorem. For odd-dimensional M we have $\chi(M) = 0$, consistent with $2e(\xi) = 0$ in odd rank.

10.3 Chern classes through the Gysin sequence

We describe the inductive construction of Chern classes used in Milnor–Stasheff, an alternative to the Grassmannian route of Lecture 6.

Theorem 10.5 (Gysin sequence). *For an oriented $\mathbb{R}^n$ -bundle ξ with total space $E \rightarrow B$ there is a long exact sequence with integer coefficients*

$$\cdots \rightarrow H^{i-n}(B) \xrightarrow{\smile e} H^i(B) \xrightarrow{\pi_0^*} H^i(E_0) \rightarrow H^{i-n+1}(B) \rightarrow \cdots$$

Proof. Start from the long exact sequence of the pair (E, E_0) and substitute $H^j(E, E_0) \cong H^{j-n}(B)$, by the Thom isomorphism, and $H^j(E) \cong H^j(B)$, by homotopy invariance. The map $H^{j-n}(B) \rightarrow H^j(B)$ so obtained is computed by

$$i^* \phi(y) = i^* (\pi^* y \smile u) = \pi^* y \smile i^* u = \pi^* (y \smile e),$$

so under the identification it becomes $\smile e$. $\square$

Construction 10.6 (Chern classes, inductively). Let $E \rightarrow B$ be a complex bundle of rank n , with its canonical orientation. Set $c_n(E) := e(E)$. Over the sphere bundle $\pi_0: S(E) = E_0 \rightarrow B$ there is a tautological complex line sub-bundle $L \subset \pi_0^*E$, spanned at a point (x, v) by v ; let $W = L^\perp$, of rank $n - 1$. In degrees $i < 2n - 1$ the Gysin sequence shows $\pi_0^*: H^i(B) \rightarrow H^i(E_0)$ is an isomorphism, so we may define

$$c_k(E) := (\pi_0^*)^{-1}c_k(W), \quad k < n,$$

inductively on the rank. The two definitions of c_k agree, as one checks on sums of line bundles using the splitting principle.

Proposition 10.7. *The total Chern class $c(E) = 1 + c_1 + \cdots + c_n$ is invertible in $H^\Pi(B; \mathbb{Z})$.*

Proof. The proof of Proposition 2.11 works verbatim over $\mathbb{Z}$: the inverse is constructed degree by degree, each coefficient being determined by the previous ones. $\square$

10.4 Conjugate bundles and Pontryagin classes

Definition 10.8. For a complex bundle ξ , the *conjugate bundle* $\bar{\xi}$ is the same real bundle equipped with the opposite complex structure $-J$.

Proposition 10.9. $c_k(\bar{\xi}) = (-1)^k c_k(\xi)$.

Proof. Strategy. Check it for line bundles, where conjugation reverses the canonical orientation and hence the sign of the Euler class, then propagate by the splitting principle and injectivity of h_n^* .

Step 1. Line bundles. For a complex line bundle L , a complex basis v of a fibre gives the oriented real basis (v, Jv) ; passing to $-J$ replaces it by $(v, -Jv)$, which has the opposite orientation. Hence $e(\bar{L}) = -e(L)$ and, since $c_1 = e$ for line bundles by Theorem 8.3, $c_1(\bar{L}) = -c_1(L)$.

Step 2. Universal case. Over $(\mathbb{CP}^\infty)^n$, conjugating $\gamma_1 \oplus \cdots \oplus \gamma_1$ conjugates each summand, so by Step 1 and the Whitney formula

$$h_n^* c_k(\bar{\gamma}_n) = \sigma_k(-a_1, \dots, -a_n) = (-1)^k \sigma_k(a_1, \dots, a_n) = (-1)^k h_n^* c_k(\gamma_n).$$

Injectivity of h_n^* gives $c_k(\bar{\gamma}_n) = (-1)^k c_k(\gamma_n)$.

Step 3. General case. If f classifies ξ then it also classifies $\bar{\xi}$ as a conjugate, in the sense that $f^* \bar{\gamma}_n \cong \bar{\xi}$; apply naturality. $\square$

Definition 10.10. For a real bundle ξ , the *complexification* is $\xi \otimes \mathbb{C}$, whose underlying real bundle is $\xi \oplus \xi$ with complex structure $J(x, y) = (-y, x)$.

Lemma 10.11. $\xi \otimes \mathbb{C} \cong \overline{\xi \otimes \mathbb{C}}$. Consequently the odd Chern classes of a complexification are 2-torsion: $2c_{2i+1}(\xi \otimes \mathbb{C}) = 0$.

Proof. The map $(x, y) \mapsto (x, -y)$ is a real bundle isomorphism $\xi \oplus \xi \rightarrow \xi \oplus \xi$, and it is conjugate linear for J :

$$J(x, -y) = (y, x), \quad \text{while} \quad -\bar{J}(x, y) \text{ corresponds to } (y, x),$$

so it is an isomorphism of complex bundles from $\xi \otimes \mathbb{C}$ onto its conjugate. By Proposition 10.9, $c_{2i+1}(\xi \otimes \mathbb{C}) = c_{2i+1}(\overline{\xi \otimes \mathbb{C}}) = -c_{2i+1}(\xi \otimes \mathbb{C})$. $\square$

Definition 10.12 (Pontryagin classes). For a real bundle $E \rightarrow B$ set

$$p_i(E) := (-1)^i c_{2i}(E \otimes \mathbb{C}) \in H^{4i}(B; \mathbb{Z}), \quad p_0 = 1,$$

and $p(E) = 1 + p_1(E) + p_2(E) + \cdots$. Naturality $f^*p_i(E) = p_i(f^*E)$ is inherited from the Chern classes.

Theorem 10.13. $2(p(E_1 \oplus E_2) - p(E_1)p(E_2)) = 0$; that is, the total Pontryagin class is multiplicative modulo 2-torsion.

Proof. Complexification is additive: $(E_1 \oplus E_2) \otimes \mathbb{C} \cong (E_1 \otimes \mathbb{C}) \oplus (E_2 \otimes \mathbb{C})$. The Whitney formula for Chern classes gives $c((E_1 \oplus E_2) \otimes \mathbb{C}) = c(E_1 \otimes \mathbb{C}) c(E_2 \otimes \mathbb{C})$. Passing from c to p discards the odd Chern classes, and every discarded term is 2-torsion by Lemma 10.11; multiplying the difference by 2 therefore kills it. $\square$

Corollary 10.14. If E is a complex bundle then $E \otimes \mathbb{C} \cong E \oplus \bar{E}$, hence

$$1 - p_1 + p_2 - \cdots + (-1)^n p_n = (1 - c_1 + c_2 - \cdots)(1 + c_1 + c_2 + \cdots),$$

so for instance $p_1 = c_1^2 - 2c_2$ and $p_2 = c_2^2 - 2c_1c_3 + 2c_4$.

Proof. For a complex E with structure J , the map

$$E \otimes \mathbb{C} = E \oplus E \longrightarrow E \oplus \bar{E}, \quad (x, y) \longmapsto \left(\frac{1}{2}(x + Jy), \frac{1}{2}(x - Jy) \right),$$

is a complex isomorphism, as one checks by applying J to both sides. Hence $c(E \otimes \mathbb{C}) = c(E) c(\bar{E})$, and Proposition 10.9 turns $c(\bar{E})$ into the alternating series. Comparing coefficients of even degree gives the stated identity. $\square$

Example 10.15. $T(S^n) \oplus \nu(S^n)$ is trivial with ν trivial, so by Theorem 10.13 and the torsion-freeness of $H^*(S^n; \mathbb{Z})$,

$$p(TS^n) = 1.$$

Pontryagin classes do not distinguish spheres from parallelizable manifolds.

In the next lecture we compute $p(T\mathbb{CP}^n)$, which is far from trivial, extract numerical invariants and put them to work on cobordism questions.

Lecture 11

Guiding question. *Time to harvest. First the model computation of the whole theory, $c(T\mathbb{CP}^n) = (1+a)^{n+1}$ and $p(T\mathbb{CP}^n) = (1+u^2)^{n+1}$. Then we turn classes into numbers and prove genuinely geometric statements: $\mathbb{CP}^{2n}$ admits no orientation reversing diffeomorphism, and it is not an oriented boundary. Finally we build the classes S_I , designed to detect products.*

11.1 The tangent bundle of complex projective space

Theorem 11.1. *Let $a = -c_1(\gamma_1 \rightarrow \mathbb{CP}^n)$, a generator of $H^2(\mathbb{CP}^n; \mathbb{Z})$. Then*

$$c(T\mathbb{CP}^n) = (1+a)^{n+1}.$$

Proof. Strategy. Repeat the argument of Lemma 2.16 over $\mathbb{C}$: a tangent direction at a line $L \subset \mathbb{C}^{n+1}$ is a complex linear map $L \rightarrow L^\perp$; adding the trivial summand $\text{Hom}_{\mathbb{C}}(\gamma_1, \gamma_1)$ produces a sum of $n+1$ copies of the dual of γ_1 , which is the conjugate bundle.

Step 1. The tangent bundle. Exactly as in the real case,

$$T\mathbb{CP}^n \cong \text{Hom}_{\mathbb{C}}(\gamma_1, \gamma_1^\perp),$$

where $\gamma_1^\perp$ is the complex orthogonal complement of γ_1 inside the trivial bundle $\mathbb{CP}^n \times \mathbb{C}^{n+1}$, taken with respect to a Hermitian metric.

Step 2. Adding a trivial line. The bundle $\text{Hom}_{\mathbb{C}}(\gamma_1, \gamma_1)$ has the nowhere zero section id , hence is the trivial complex line bundle $\varepsilon_{\mathbb{C}}^1$. Therefore

$$T\mathbb{CP}^n \oplus \varepsilon_{\mathbb{C}}^1 \cong \text{Hom}_{\mathbb{C}}(\gamma_1, \gamma_1^\perp \oplus \gamma_1) = \text{Hom}_{\mathbb{C}}(\gamma_1, \mathbb{C}^{n+1}) \cong (\gamma_1^*)^{\oplus(n+1)}.$$

Step 3. The dual is the conjugate. A Hermitian metric h identifies $\gamma_1^* = \text{Hom}_{\mathbb{C}}(\gamma_1, \mathbb{C})$ with $\overline{\gamma_1}$: the map $v \mapsto h(\cdot, v)$ is conjugate linear in v , hence complex linear as a map into the conjugate bundle. By Proposition 10.9, $c_1(\overline{\gamma_1}) = -c_1(\gamma_1) = a$.

Step 4. Conclusion. Using Step 2, the Whitney formula and Step 3,

$$c(T\mathbb{CP}^n) = c(T\mathbb{CP}^n \oplus \varepsilon_{\mathbb{C}}^1) = c(\overline{\gamma_1})^{n+1} = (1+a)^{n+1}.$$

□

Theorem 11.2. *With $u = a$ as above,*

$$p(T\mathbb{CP}^n) = (1 + u^2)^{n+1}, \quad \text{i.e.} \quad p_k(T\mathbb{CP}^n) = \binom{n+1}{k} u^{2k}.$$

Proof. By Corollary 10.14 applied to $E = T\mathbb{CP}^n$,

$$1 - p_1 + p_2 - \cdots = c(T\mathbb{CP}^n) c(\overline{T\mathbb{CP}^n}) = (1 + u)^{n+1} (1 - u)^{n+1} = (1 - u^2)^{n+1},$$

where the middle equality uses Theorem 11.1 and Proposition 10.9. Comparing the coefficients of u^{2k} on both sides, the sign $(-1)^k$ occurs on each, so $p_k = \binom{n+1}{k} u^{2k}$. $\square$

Example 11.3. $p(T\mathbb{CP}^5) = (1 + u^2)^6 = 1 + 6u^2 + 15u^4$, the higher terms vanishing in $H^*(\mathbb{CP}^5; \mathbb{Z})$. In contrast with spheres, Pontryagin classes see projective spaces very clearly.

Remark 11.4. For an oriented real bundle V of rank n there are two natural orientations on $V \otimes \mathbb{C} \cong V \oplus V$: the complex one, given by $v_1, Jv_1, \dots, v_n, Jv_n$, and the direct sum one, given by $v_1, \dots, v_n, Jv_1, \dots, Jv_n$. Reordering one into the other requires $n(n-1)/2$ transpositions, so the two differ by the sign $(-1)^{n(n-1)/2}$.

Lemma 11.5. *If V is an oriented real bundle of rank $2n$ then $p_n(V) = e(V)^2$.*

Proof. By definition $p_n(V) = (-1)^n c_{2n}(V \otimes \mathbb{C})$, and $c_{2n} = e$ for a complex bundle of rank $2n$ by Theorem 8.3, computed with the complex orientation. By Remark 11.4 with n replaced by $2n$, the complex and sum orientations differ by $(-1)^{2n(2n-1)/2} = (-1)^n$, so

$$p_n(V) = (-1)^n \cdot (-1)^n e(V \oplus V) = e(V)^2,$$

using $e(V \oplus V) = e(V) \smile e(V)$ from Theorem 7.12(5). $\square$

11.2 Chern numbers and Pontryagin numbers

Definition 11.6. Let X be a closed almost complex manifold of complex dimension n , so that TX carries a complex structure, and let $I = (i_1, \dots, i_r)$ be a partition of n . The *Chern number* is

$$C_I(X) = \langle c_{i_1}(TX) \smile \cdots \smile c_{i_r}(TX), [X] \rangle \in \mathbb{Z},$$

where $[X]$ is the fundamental class of the complex orientation.

Definition 11.7. Let M^{4n} be closed and oriented and $I = (i_1, \dots, i_r)$ a partition of n . The *Pontryagin number* is

$$P_I(M) = \langle p_{i_1}(TM) \cdots p_{i_r}(TM), [M] \rangle \in \mathbb{Z}.$$

Example 11.8. By Theorem 11.1, $C_I(\mathbb{CP}^n) = \prod_j \binom{n+1}{i_j} > 0$ for every partition I of n ; and by Theorem 11.2, $P_I(\mathbb{CP}^{2n}) = \prod_j \binom{2n+1}{i_j} \neq 0$ for every partition I of n .

Remark 11.9. For an oriented bundle V and its reverse $\bar{V}$ one has $p_i(V) = p_i(\bar{V})$, since the complexification does not see the orientation. Reversing the orientation of the *manifold*, however, reverses $[M]$ and therefore changes the sign of every Pontryagin number.

Lemma 11.10. *If some Pontryagin number of M^{4n} is non-zero, then M admits no orientation reversing diffeomorphism.*

Proof. Let $f: M \rightarrow M$ be a diffeomorphism. Its differential is a bundle isomorphism $TM \cong f^*TM$, so $f^*p_i(TM) = p_i(f^*TM) = p_i(TM)$. If moreover f reverses orientation then $f_*[M] = -[M]$, whence for every I

$$P_I(M) = \langle p_I(TM), [M] \rangle = \langle f^*p_I(TM), [M] \rangle = \langle p_I(TM), f_*[M] \rangle = -P_I(M),$$

forcing all Pontryagin numbers to vanish. $\square$

Corollary 11.11. $\mathbb{CP}^{2n}$ admits no orientation reversing diffeomorphism. By contrast $\mathbb{CP}^{2n+1}$ does, since its Pontryagin numbers live in the wrong dimension to obstruct anything.

Lemma 11.12. *If M^{4n} is the boundary of a compact oriented $(4n+1)$ -manifold B , then all Pontryagin numbers of M vanish.*

Proof. Strategy. The same argument as Theorem 3.2, now with integral coefficients, which is why orientations are needed: the normal bundle of the boundary is canonically oriented by the outward direction.

Step 1. Along $M = \partial B$ the outward pointing direction trivializes the normal bundle, so $TB|_M \cong TM \oplus \varepsilon^1$ as oriented bundles and $i^*p_j(TB) = p_j(TM)$ for the inclusion $i: M \hookrightarrow B$.

Step 2. With $\mathbb{Z}$ coefficients, $\partial[B] = [M]$ for the fundamental class $[B] \in H_{4n+1}(B, M; \mathbb{Z})$ of the compact oriented manifold with boundary.

Step 3. For $v = p_{i_1}(TB) \cdots p_{i_r}(TB) \in H^{4n}(B; \mathbb{Z})$,

$$P_I(M) = \langle i^*v, [M] \rangle = \langle i^*v, \partial[B] \rangle = \langle \delta i^*v, [B] \rangle = 0,$$

since $\delta \circ i^* = 0$ by exactness of the sequence of the pair (B, M) . $\square$

Example 11.13. $\mathbb{CP}^{2n}$ is not an oriented boundary. On the other hand $\mathbb{CP}^{2n} \sqcup (-\mathbb{CP}^{2n}) = \partial(\mathbb{CP}^{2n} \times [0, 1])$, so in the oriented cobordism group $[\mathbb{CP}^{2n}] +$

$[-\mathbb{CP}^{2n}] = 0$, while $2[\mathbb{CP}^{2n}] \neq 0$: by Lemma 11.10 the two orientations of $\mathbb{CP}^{2n}$ really are different.

11.3 The symmetric function classes S_I

Guiding question. *Chern numbers of a product mix the classes of the two factors in a complicated way. Is there a family of characteristic classes adapted to sums and products, one that vanishes on non-trivial products?*

For partitions $J = (j_1, \dots, j_s)$ and $K = (k_1, \dots, k_t)$ write JK for the concatenated partition $(j_1, \dots, j_s, k_1, \dots, k_t)$.

Definition 11.14. For a partition $I = (i_1, \dots, i_r)$ let

$$S_I(u_1, \dots, u_n) = \sum u_{\alpha_1}^{i_1} \cdots u_{\alpha_r}^{i_r}$$

be the smallest symmetric polynomial containing the monomial $u_1^{i_1} \cdots u_r^{i_r}$, each distinct monomial appearing exactly once. Being symmetric, it is a polynomial in $\sigma_1, \dots, \sigma_n$, and for a complex bundle E we set

$$S_I(E) := S_I(c_1(E), \dots, c_n(E)) \in H^{2|I|}(B).$$

Equivalently $S_I(E) = f^* S_I(\tilde{\sigma}_1, \dots, \tilde{\sigma}_n)$ for a classifying map f . For instance $S_{(1)} = \sigma_1$, $S_{(1,1)} = \sigma_2$, and $S_{(2)} = \sigma_1^2 - 2\sigma_2$, so $S_{(2)}(E) = c_1^2 - 2c_2$.

Theorem 11.15. *For complex bundles E, E' over the same base,*

$$S_I(E \oplus E') = \sum_{JK=I} S_J(E) \smile S_K(E').$$

Proof. Strategy. Verify the identity for sums of line bundles, where the Chern roots of $E \oplus E'$ are the union of those of E and E' ; the combinatorics is the distribution of the exponents of I between the two groups of roots. The splitting principle then gives the general case.

Step 1. Reduction to roots. By the splitting principle (Remark 6.12) there is a map f , injective on cohomology, such that f^*E and f^*E' split into line bundles, with Chern roots $u'_1, \dots, u'_n$ and $u''_1, \dots, u''_n$. The Chern roots of the sum are all of these together, so it suffices to prove the corresponding polynomial identity.

Step 2. Distributing the exponents. By definition, S_I evaluated on the combined variables is the sum of all distinct monomials $v_{\alpha_1}^{i_1} \cdots v_{\alpha_r}^{i_r}$, where each v_α is one of the u' or one of the u'' and the indices are distinct. Group the terms according to which exponents are assigned to primed variables: this is exactly a splitting of the partition I into a sub-partition J used on the u' and the complementary K used on the u'' . Summing over all such splittings gives

$$S_I(u', u'') = \sum_{JK=I} S_J(u') S_K(u''),$$

each monomial being counted once on both sides.

Step 3. Back to bundles. Applying f^* and using its injectivity gives the identity for E, E' . $\square$

Corollary 11.16. *For a partition with a single part, $S_{(m)}(E \oplus E') = S_{(m)}(E) + S_{(m)}(E')$, since the only splittings of (m) are $J = (m), K = \emptyset$ and the reverse.*

Definition 11.17. For a closed complex manifold X of complex dimension m put

$$S_m[X] := \langle S_{(m)}(c(TX)), [X] \rangle \in \mathbb{Z}.$$

Corollary 11.18. *If K^m and L^n are closed complex manifolds of positive dimension then $S_{m+n}[K \times L] = 0$.*

Proof. Since $T(K \times L) = p_1^*TK \oplus p_2^*TL$, Corollary 11.16 gives

$$S_{(m+n)}(T(K \times L)) = p_1^*S_{(m+n)}(TK) + p_2^*S_{(m+n)}(TL).$$

The first summand lies in the image of $H^{2(m+n)}(K)$, which vanishes because K has complex dimension $m < m+n$; likewise for the second. Hence the class itself is zero. $\square$

Example 11.19. $S_n[\mathbb{CP}^n] = n+1 \neq 0$, because $T\mathbb{CP}^n \oplus \varepsilon^1$ has $n+1$ Chern roots all equal to a , so $S_{(n)} = \sum a^n = (n+1)a^n$. By Corollary 11.18, $\mathbb{CP}^n$ is not a non-trivial product of complex manifolds.

Theorem 11.20. *Let $X^1, \dots, X^n$ be closed complex manifolds with $\dim_{\mathbb{C}} X^k = k$ and $S_k[X^k] \neq 0$ for every k . Then the $p(n) \times p(n)$ matrix of Chern numbers*

$$\left(C_I(X^{j_1} \times \dots \times X^{j_s}) \right)_{I, J \vdash n}$$

is non-degenerate, where $p(n)$ is the number of partitions of n and $J = (j_1, \dots, j_s)$ runs over them.

The proof is a triangularity argument with respect to a suitable order on partitions; it opens the final lecture, where the same result in its Pontryagin form feeds directly into the computation of the oriented cobordism ring.

Lecture 12

Guiding question. *Everything converges here. The S_I matrix theorem shows that products of projective spaces are independent in cobordism; the oriented cobordism ring organizes the question; and Thom's theorem converts cobordism into homotopy theory. Together they compute $\Omega_* \otimes \mathbb{Q}$.*

12.1 Proof of the matrix theorem

Proof of Theorem 11.20. Strategy. Order partitions by their number of parts. The multiplicativity of S_I shows that S_I can be non-zero on a product indexed by J only when I has at least as many parts as J , and that on the diagonal $I = J$ it is a product of non-zero numbers. That makes the matrix of the S_I triangular with non-zero diagonal; a change of basis converts it into the matrix of Chern numbers.

Step 1. Multiplicativity. Iterating Theorem 11.15 across a product $X^{j_1} \times \cdots \times X^{j_s}$, whose tangent bundle is the sum of the pull-backs of the TX^{j_t} ,

$$S_I(X^{j_1} \times \cdots \times X^{j_s}) = \sum_{I_1 I_2 \cdots I_s = I} S_{I_1}(X^{j_1}) \cdots S_{I_s}(X^{j_s}),$$

the sum being over all ways of splitting I into s consecutive sub-partitions.

Step 2. Which terms survive evaluation. Evaluating on the fundamental class, a term vanishes unless each I_t has weight j_t , that is, unless I_t is a partition of j_t : otherwise the corresponding factor lies in a cohomology group above or below the top degree of X^{j_t} . In particular each I_t is non-empty, so

$$S_I[X^{j_1} \times \cdots \times X^{j_s}] \neq 0 \implies \#\{\text{parts of } I\} \geq \#\{\text{parts of } J\}.$$

Step 3. The diagonal. If $I = J$ as partitions, the only splittings that contribute are those with $I_t = (j_t)$ a single part, so

$$S_J[X^{j_1} \times \cdots \times X^{j_s}] = \lambda_J \prod_{t=1}^s S_{j_t}[X^{j_t}],$$

where λ_J is a positive integer counting the number of such splittings. By hypothesis every factor is non-zero, so the diagonal entry is non-zero.

Step 4. Triangularity. Order the $p(n)$ partitions of n so that the number of parts is non-decreasing. By Step 2 the matrix $(S_I[X^J])_{I,J}$ vanishes below the diagonal blocks and by Step 3 has non-zero entries on the diagonal, so its determinant is non-zero.

Step 5. From S_I to Chern numbers. Each S_I is an integral linear combination of the monomials $c_{i_1} \cdots c_{i_r}$ of the same weight, and conversely, because both families are bases of the same lattice, as shown in the next paragraph. Hence the matrix of Chern numbers is obtained from the matrix of the S_I by an invertible change of basis and is therefore also non-degenerate. $\square$

The change of basis. Inside $\mathbb{Z}[u_1, \dots, u_n]$ the symmetric polynomials form $\mathbb{Z}[\sigma_1, \dots, \sigma_n]$. Its part of degree k has two natural bases indexed by the partitions $\lambda \vdash k$ into at most n parts:

- the monomials in the elementary symmetric functions, $\sigma_\lambda = \sigma_{\lambda_1} \sigma_{\lambda_2} \cdots$, equivalently $\sigma_1^{r_1} \cdots \sigma_n^{r_n}$ with $\sum i r_i = k$;
- the symmetrized monomials S_λ , each distinct monomial of exponent pattern λ appearing once.

The first is a basis by the fundamental theorem of symmetric polynomials; the second because every symmetric polynomial is by definition a sum of symmetrized monomials. The transition matrix between two bases of the same free module is invertible over $\mathbb{Z}$, which is what Step 5 used.

Remark 12.1. When applying Theorem 11.15 it matters that each monomial is counted once. Writing S_I as a sum over all of S_n would count each monomial $|\text{Iso}|$ times, where Iso is the isotropy group of the exponent pattern; the multiplicativity formula holds for the normalized version.

The Pontryagin version. For a closed oriented M^{4m} and $I \vdash m$ define $S_I(p(E))$ by the same universal polynomial in the Pontryagin classes and put $S_I[M] = \langle S_I(p(TM)), [M] \rangle$. Since the Pontryagin classes are multiplicative only modulo 2-torsion (Theorem 10.13),

$$S_I(p(E \oplus E')) = \sum_{JK=I} S_J(p(E)) S_K(p(E')) \quad \text{modulo 2-torsion,}$$

which is enough for evaluation against fundamental classes, torsion classes pairing to zero with an integral class after multiplication by 2.

Theorem 12.2. *Let $\{M^{4k}\}_{k \geq 1}$ be closed oriented manifolds with $S_k[M^{4k}] \neq 0$. Then for every n the $p(n) \times p(n)$ matrix of Pontryagin numbers*

$$\left(P_I(M^{4j_1} \times \cdots \times M^{4j_s}) \right)_{I, J \vdash n}$$

is non-degenerate. The family $M^{4k} = \mathbb{CP}^{2k}$ qualifies, since $p(T\mathbb{CP}^{2k}) = (1 + u^2)^{2k+1}$ gives $S_k[\mathbb{CP}^{2k}] = 2k + 1 \neq 0$.

Proof. The five steps of the proof above apply verbatim, with c replaced by p and degrees multiplied by two; the only change is that the identities hold modulo 2-torsion, which does not affect the evaluations. $\square$

Remark 12.3 (quoted). The classifying space for *oriented* bundles is the oriented Grassmannian $\tilde{G}_n$ of oriented n -planes, and

$$H^*(\tilde{G}_{2m+1}; \mathbb{Q}) = \mathbb{Q}[p_1, \dots, p_m],$$

$$H^*(\tilde{G}_{2m}; \mathbb{Q}) = \mathbb{Q}[p_1, \dots, p_{m-1}, e], \quad e^2 = p_m.$$

So rationally the Pontryagin classes, together with the Euler class in even rank, are all the characteristic classes of oriented bundles.

12.2 The oriented cobordism ring

Definition 12.4. Closed oriented manifolds M, M' of dimension n are *oriented cobordant* if there is a compact oriented X together with an orientation preserving diffeomorphism $M \sqcup (-M') \cong \partial X$, the boundary carrying its induced orientation. Let Ω_n be the set of equivalence classes.

Proposition 12.5. Ω_n is an abelian group under disjoint union, with zero the class of boundaries and with $-[M] = [-M]$; and $\Omega_* = \bigoplus_{n \geq 0} \Omega_n$ is a graded commutative ring under Cartesian product.

Proof. Reflexivity uses $M \sqcup (-M) = \partial(M \times [0, 1])$; symmetry is reversal of orientation of the cobordism; transitivity is gluing two cobordisms along a common boundary component, which is again a smooth manifold after smoothing the corner. Associativity and commutativity of $\sqcup$ and $\times$ are clear, and the product of a boundary with a closed manifold is a boundary, so the product descends to classes. $\square$

Lemma 12.6. *Oriented cobordant manifolds have equal Pontryagin numbers.*

Proof. If $M \sqcup (-M')$ bounds, Lemma 11.12 gives $P_I(M) + P_I(-M') = 0$, and $P_I(-M') = -P_I(M')$. $\square$

Corollary 12.7. *For every partition I of k the assignment $[M] \mapsto P_I(M)$ is a group homomorphism $\Omega_{4k} \rightarrow \mathbb{Z}$. The products $\mathbb{CP}^{2i_1} \times \dots \times \mathbb{CP}^{2i_r}$, as $I = (i_1, \dots, i_r)$ runs over the partitions of k , are linearly independent in Ω_{4k} , so*

$$\text{rk } \Omega_{4k} \geq p(k).$$

Proof. Assemble the homomorphisms into $\Omega_{4k} \rightarrow \mathbb{Z}^{p(k)}$, $[M] \mapsto (P_I(M))_I$. By Theorem 12.2 the images of the $p(k)$ products $\mathbb{CP}^{2i_1} \times \dots \times \mathbb{CP}^{2i_r}$ form a

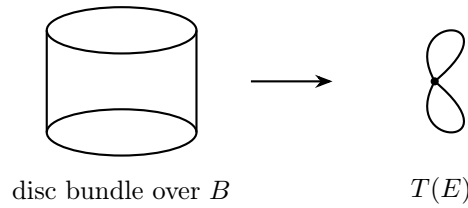
non-degenerate matrix, hence are linearly independent over $\mathbb{Q}$; so the classes themselves are independent. $\square$

12.3 Thom spaces

Definition 12.8. Let $\pi: E \rightarrow B$ be a real bundle of rank k with a Euclidean metric, and $E^{\geq 1} = \{e \mid |e| \geq 1\}$. The *Thom space* is the pointed space

$$T(E) = E/E^{\geq 1}, \quad *_E = [E^{\geq 1}],$$

that is, the unit disc bundle with its boundary sphere bundle collapsed to a point.



Proposition 12.9. $\tilde{H}^{i+k}(T(E)) \cong H^i(B)$ by the *Thom isomorphism*, for oriented E over $\mathbb{Z}$. If B is a CW complex then $T(E)$ is a CW complex with one $(d+k)$ -cell for each d -cell of B , plus the basepoint; in particular $T(E)$ is $(k-1)$ -connected.

Proof. Since $E^{\geq 1}$ is a deformation retract of E_0 , we have

$$\tilde{H}^*(T(E)) \cong H^*(E, E^{\geq 1}) \cong H^*(E, E_0) \cong H^{*-k}(B)$$

by Theorem 7.7. For the cell structure, over a d -cell D^d the bundle is trivial, so the part of the disc bundle above it is $D^d \times D^k$, contributing one $(d+k)$ -cell after the collapse. Since $T(E)$ then has no cells in dimensions $1, \dots, k-1$ apart from the basepoint, a cellular approximation argument shows every map $S^m \rightarrow T(E)$ with $m < k$ is null-homotopic. $\square$

12.4 Thom's theorem

Definition 12.10. Let $\mathcal{C}$ be the class of finite abelian groups. A homomorphism $h: A \rightarrow B$ is a $\mathcal{C}$ -isomorphism if $\ker h$ and $\operatorname{coker} h$ both lie in $\mathcal{C}$.

Theorem 12.11 (Hurewicz modulo $\mathcal{C}$, quoted). *If X is a finite CW complex which is $(k-1)$ -connected, then the Hurewicz map $\pi_r(X) \rightarrow H_r(X; \mathbb{Z})$ is a $\mathcal{C}$ -isomorphism for $r < 2k-1$.*

Corollary 12.12. *For an oriented rank k bundle E over a finite CW complex B , and $n < k - 1$, the composite*

$$\pi_{n+k}(T(E)) \longrightarrow H_{n+k}(T(E); \mathbb{Z}) \cong H_n(B; \mathbb{Z})$$

is a $\mathcal{C}$ -isomorphism.

Theorem 12.13 (Thom). *For $k > n + 1$ there is a canonical isomorphism*

$$\pi_{n+k}(T(\tilde{\gamma}^k)) \cong \Omega_n,$$

where $\tilde{\gamma}^k$ is the universal oriented k -plane bundle over $\tilde{G}_k$.

Proof of surjectivity. Strategy. Given a closed oriented M^n , embed it in a sphere, and collapse everything outside a tubular neighbourhood; what remains is a map from the sphere to the Thom space of the normal bundle, which maps to the Thom space of the universal bundle. The preimage of the zero section is M by construction.

Step 1. Embedding. By Whitney, embed $M^n \subset \mathbb{R}^{n+k} \subset S^{n+k}$ and let E be its normal bundle, oriented by $TM \oplus E = T\mathbb{R}^{n+k}|_M$.

Step 2. The collapse map. Let $U \cong E$ be a tubular neighbourhood (Theorem 9.1). Collapsing the complement of U gives

$$S^{n+k} \longrightarrow S^{n+k}/(S^{n+k} \setminus U) \cong T(E),$$

a continuous map sending everything outside the tube to the basepoint.

Step 3. To the universal Thom space. A bundle map $E \rightarrow \tilde{\gamma}^k$, which exists by the oriented version of Theorem 4.5, induces a pointed map $T(E) \rightarrow T(\tilde{\gamma}^k)$. Composing with Step 2 gives $g: S^{n+k} \rightarrow T(\tilde{\gamma}^k)$.

Step 4. Recovering M . By construction g is smooth and transverse to the zero section near the preimage of the zero section, and $g^{-1}(\text{zero section}) = M$ with normal bundle $g^*\tilde{\gamma}^k \cong E$. Hence the class of g maps to $[M]$. $\square$

Remark 12.14 (quoted). For injectivity, and to define the inverse map, one uses transversality: every continuous $g: S^{n+k} \rightarrow T(E)$ is homotopic to one which is smooth and transverse to the zero section on the preimage of $E^{\leq 1}$, and then $g^{-1}(\text{zero section})$ is a closed oriented n -manifold whose cobordism class depends only on the homotopy class of g , a homotopy made transverse providing the cobordism. We do not prove the transversality statements.

Corollary 12.15. *For k large,*

$$\Omega_n \otimes \mathbb{Q} \cong \pi_{n+k}(T(\tilde{\gamma}^k)) \otimes \mathbb{Q} \cong H_n(\tilde{G}_k; \mathbb{Q}) = \begin{cases} \mathbb{Q}^{p(m)} & n = 4m, \\ 0 & \text{otherwise,} \end{cases}$$

using Corollary 12.12, which becomes an isomorphism after tensoring with $\mathbb{Q}$ since finite groups die, together with the rational cohomology of $\tilde{G}_k$ quoted above. Combining with Corollary 12.7, where the lower bound $p(m)$ is realized by products of even complex projective spaces,

$$\Omega_* \otimes \mathbb{Q} = \mathbb{Q}[\mathbb{CP}^2, \mathbb{CP}^4, \mathbb{CP}^6, \dots].$$

Remark 12.16 (where this leads). Everything in the course feeds forward. Hirzebruch's signature theorem expresses the signature of M^{4k} as a universal polynomial in the Pontryagin classes, precisely because both sides are cobordism invariants and agree on the generators $\mathbb{CP}^{2i}$. Milnor's exotic spheres are detected by the failure of an identity between Pontryagin numbers. And index theory reinterprets all of these numbers analytically.